NARASARAOPETA ENGINEERING COLLEGE
(AUTONOMOUS)

Department of Artificial Intelligence & Machine Learning

II Year II Semester

DESIGN THINKING & INNOVATION LAB (R23CC22L3)

Roll Number: ________________

Name of the Student: ______________________

Academic Year: 2024-2025

NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

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NARASARAOPETA ENGINEERING COLLEGE
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DEPARTMENT OF

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

CERTIFICATE

ACADEMIC YEAR: 2024-25

This is to certify that the bonafide record work done by______________________ ___________________________________bearing **Roll Number** _______________ of **II B.Tech. II Semester** in the **Design Thinking & Innovation Laboratory** is satisfactorily completed.

Staff In-Charge **Head of the Department**

Submitted for the practical examination held on ______________________

External Examiner

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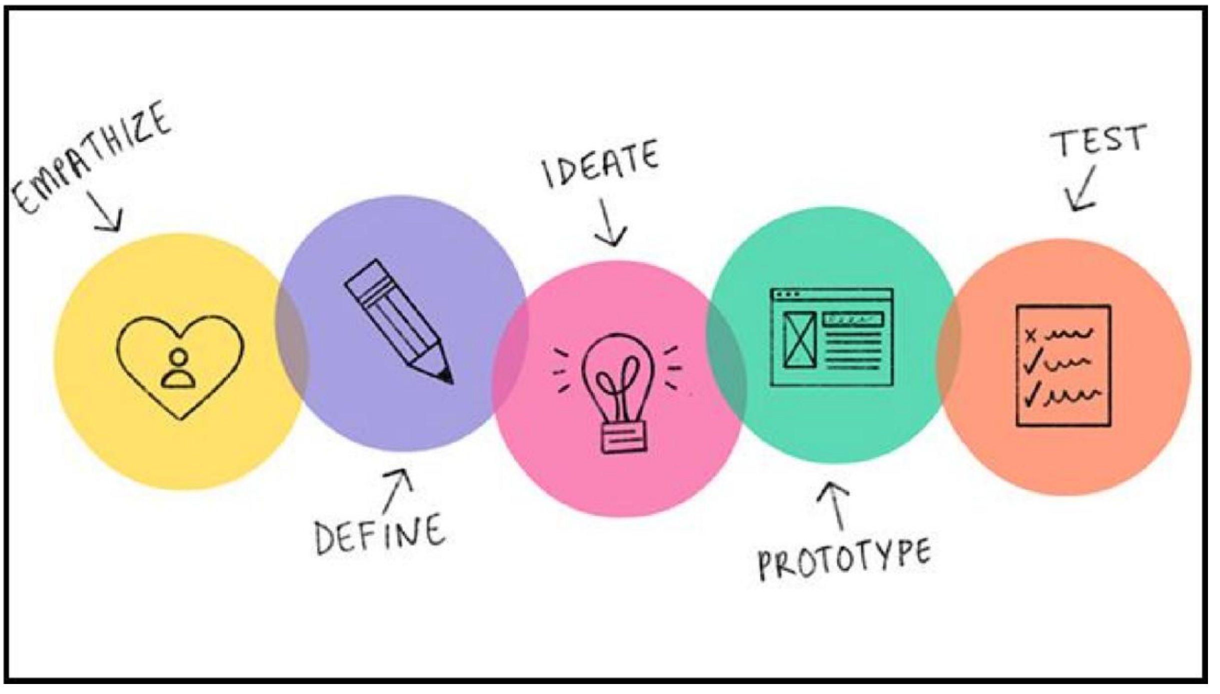
Experiment-1

AIM

Design a mind map of design thinking

Description

The **design thinking process** is a creative one that welcomes ambiguity and uncertainty. It is about taking a holistic view of a problem and considering all aspects of it. Design mindset applied to a life situation aids the people concerned in looking at the bigger picture, being informative, and acting accordingly.



Steps of Design Thinking Process

The **design thinking process** incorporates five steps:

Empathize: In this stage, you or the team understands the perspective of the target audience, consumer, or customer. You identify or address a problem that exists. Design thinkers are thus encouraged to put aside assumptions and objectively consider possibilities about the target customers and the needs they wish to address. It is usually done by observations and qualitative interviews.

Tools such as empathy maps are used to consolidate information and capture what people think, say, and feel about a problem or need.

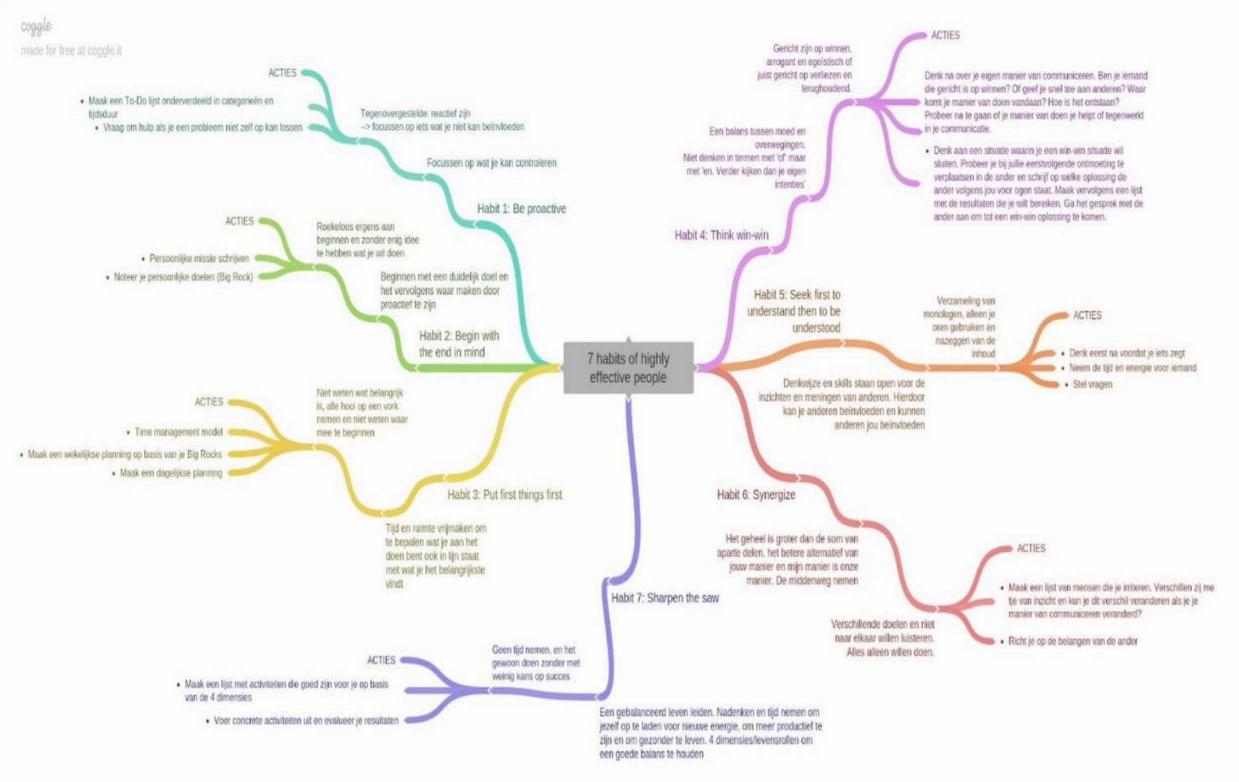
Define: At this stage, the team puts together information gathered in the first stage. They define problem statements that should be in human-centric terms rather than as business targets. For instance, the target for a pre-cooked meal product could be to benefit busy moms who have meals ready for their families in no time.

Ideate: When a problem is defined, the team can then brainstorm as to how the needs can be addressed. The team puts together all possible ideas and then starts to investigate or test them. It is a creative, freeing phase that allows the team to think out of the box.

Prototype: The team identifies which are possible solutions for the problem identified. It is a scaled-down version of systems and products which they can present to the target customer group and get their feedback as well.

The Anatomy of a Mind Map:

Mind maps always start from a central point, which is the main topic, and branch out into subcomponents. Here's an example of a simple mind map:

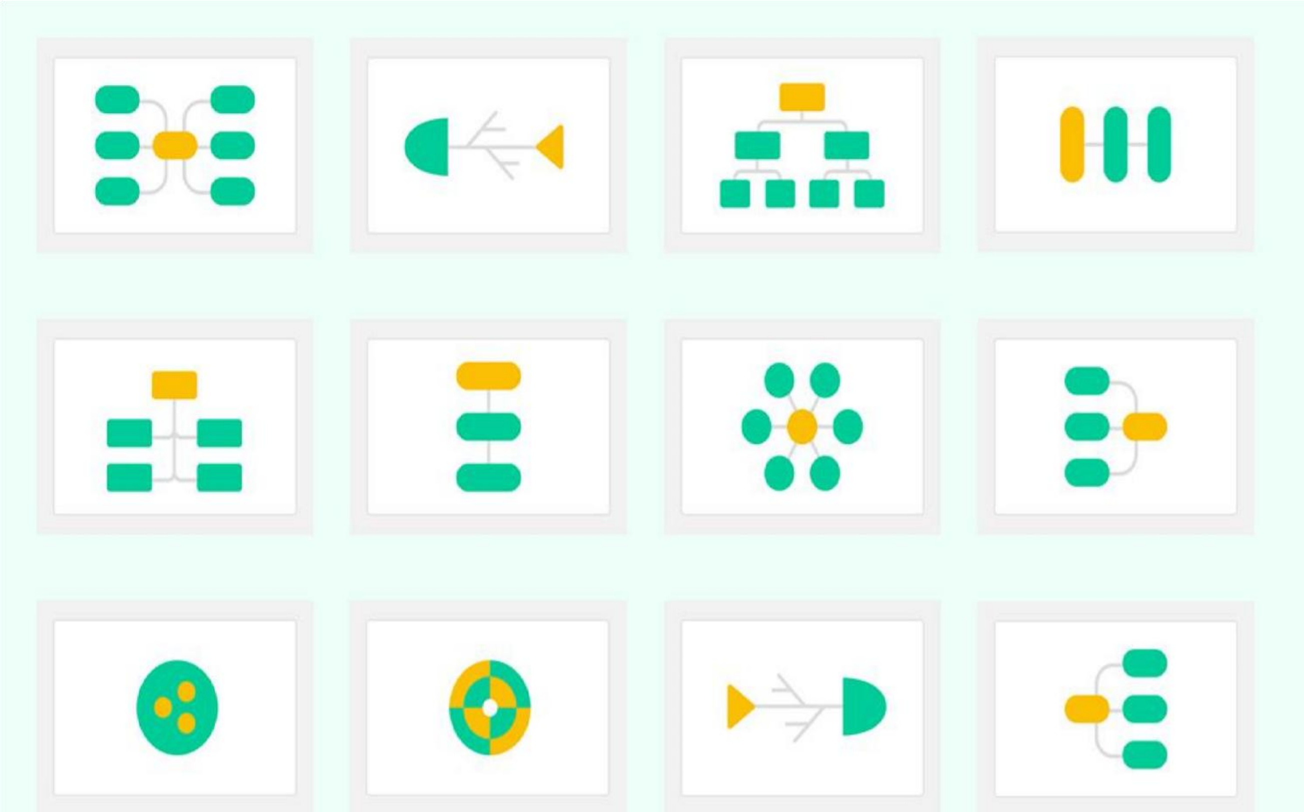


How to Do Design Thinking Process with Mind Maps

There are several steps in the design thinking process, and each of them requires ideas and feedback to be collated and brainstormed over. They need to be shared between team members, presented to target groups, and solutions to be mapped to the problem.

12 Mind Map Structures

The software provides 12 structure of mind maps, 33 themes, and 700 plus clip chart images. You can create empathy maps or ideas of a brainstorming session in the form of radial mind maps, bubble or timeline, sector, or fishbone maps.



Mind Mapping Technique Tips

Here are some tips for mind mapping in design thinking:

- A short map can provide a wealth of information.
- Multitasking of maps on the same canvas on demand of user needs.
- By indicating hierarchy on the map, designers can create wireframes much more quickly.
- Don't limit your brainstorming; collect useful and undesirable ideas, then hone the best ones.

What Are the Benefits of Using Mind Mapping?

We can understand concepts using mind maps in the design process.

- Growth of cognitive designing thinking
- Helps in targeting the audience for the growth of the business
- Untangling the problems and coming up with a defined solution

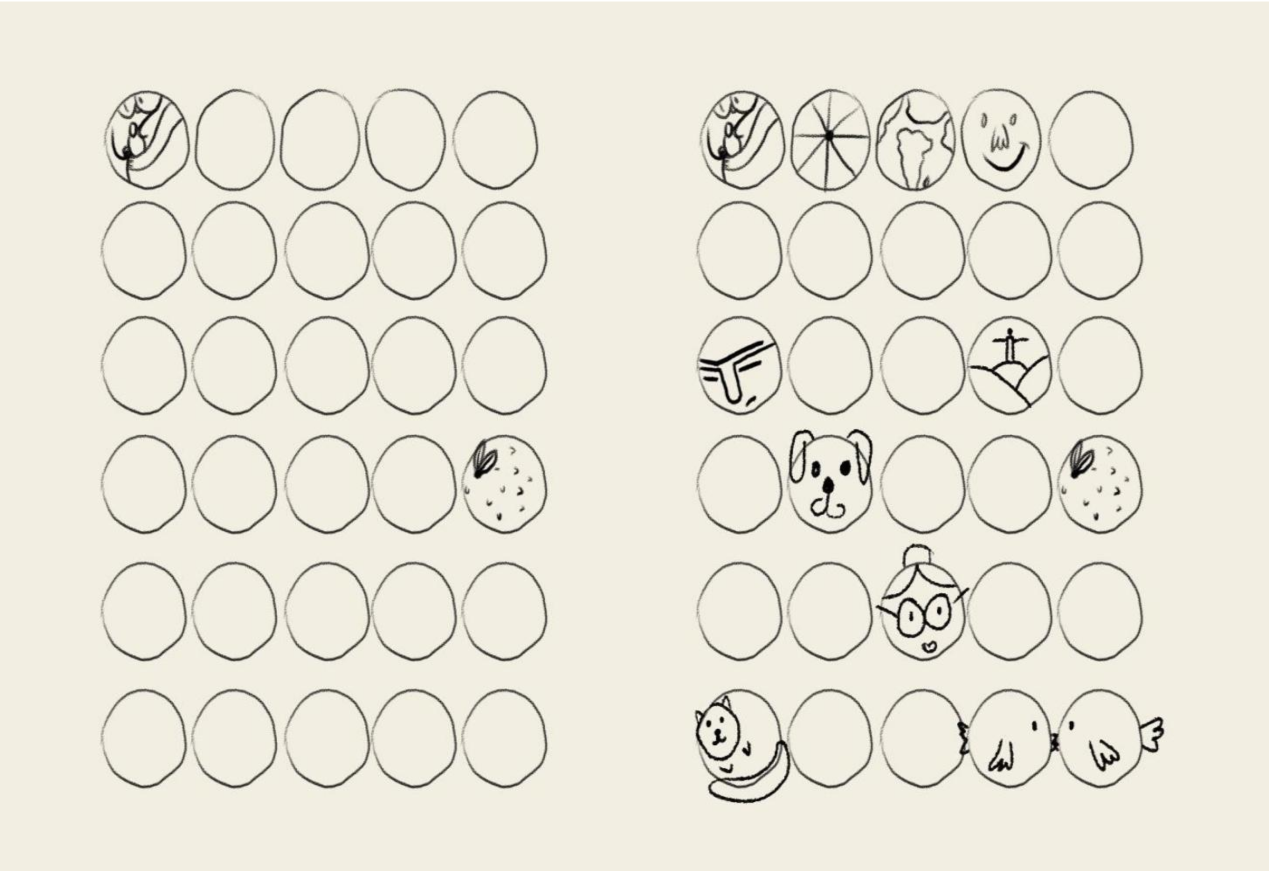
- Designers can understand the apps and websites according to users' perspectives
- For the user, app, and designer, it functions like a prism

Experiment-2

AIM

Thirty circle Exercise ---ideation

Description



The 'Thirty Circles' Exercise

<u>Creativity</u>

The mind is like a muscle. The more you use it, the stronger it gets. This statement is as true for creativity as it is for anything else.

One exercise to get your inventive brain warmed up is the ‘Thirty Circles’ exercise. Don’t panic, this is an unbelievably simple practice and you can do it on your own or in a group.

Step one is to draw or print out thirty, same-sized circles on a large piece of paper - A4 is perfect.

(If you're drawing them, you could use the bottom of a shot glass to trace out the circles).

Step two is to take a pen and turn each circle into something using your imagination. Give yourself no more than 3 minutes to complete the task.

For example, one circle could be an image of planet earth. Another could be an orange. Or a bicycle wheel. Or even a basketball. You get the idea.

The goal of the exercise is to get your brain thinking broadly and is a form of divergent thinking.

To give you some ideas, have a look at the image above. There is also a variant of this exercise using two circles instead of one. In this instance, you might imagine the two wheels of a bicycle or a pair of glasses, for example.

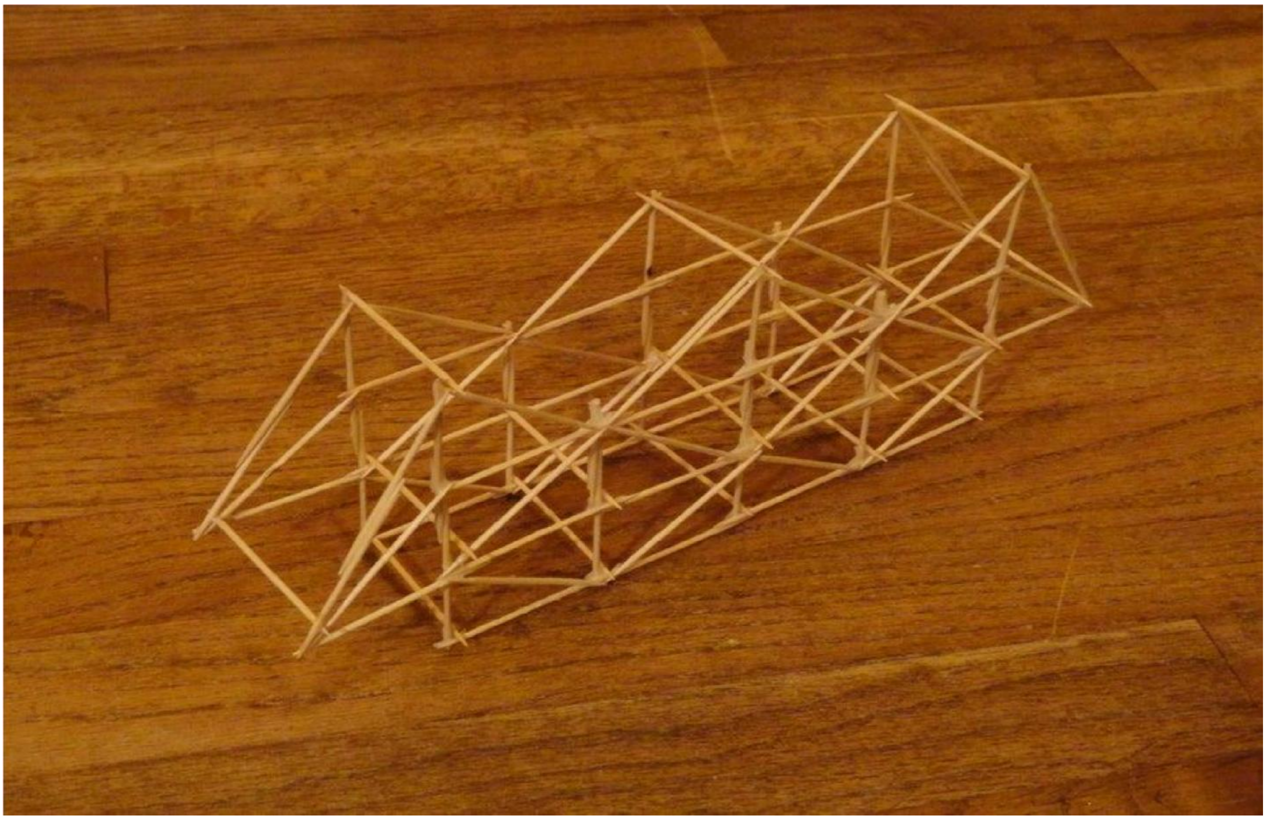
Once you've completed the exercise, reflect on the number and diversity of ideas. What did you learn? Creative warmup exercises like this one are a great way to lower your inhibitions and avoid being critical of yourself before you've got started.

Experiment-3

AIM

Prepared a toothpick bridge (mock-up model)

Description



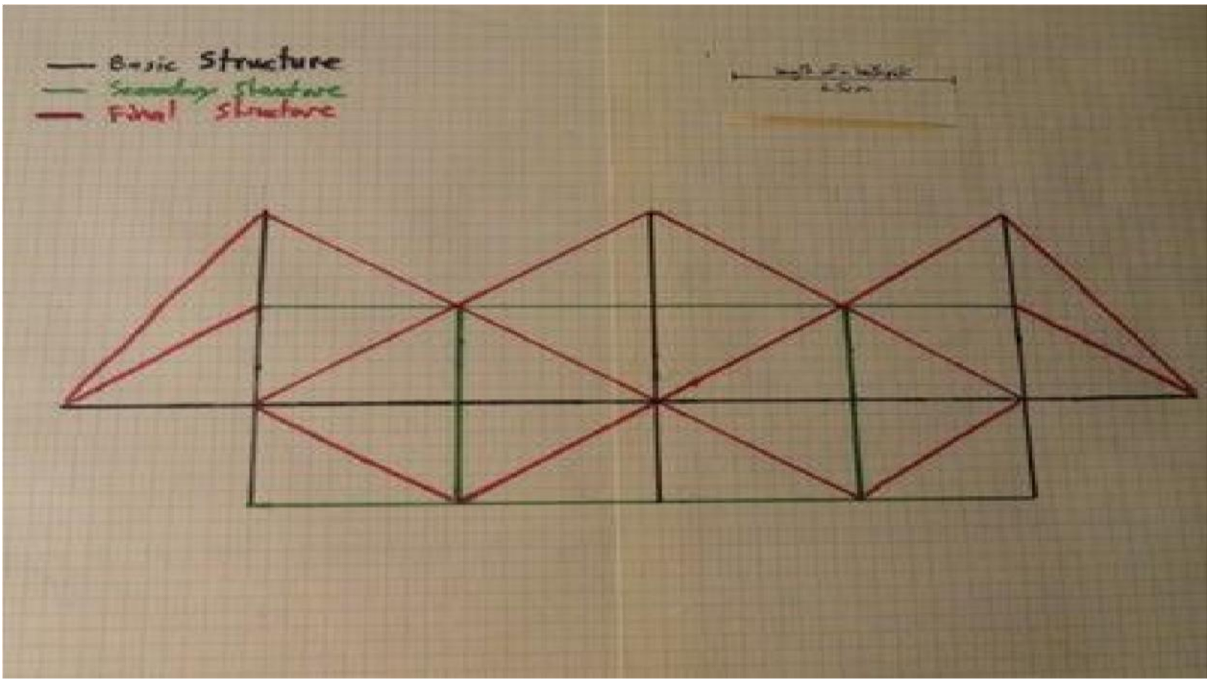
This is a cheap and easy project that only requires a small amount of materials. The Toothpick Bridge requires a lot of thinking and a good amount of construction.

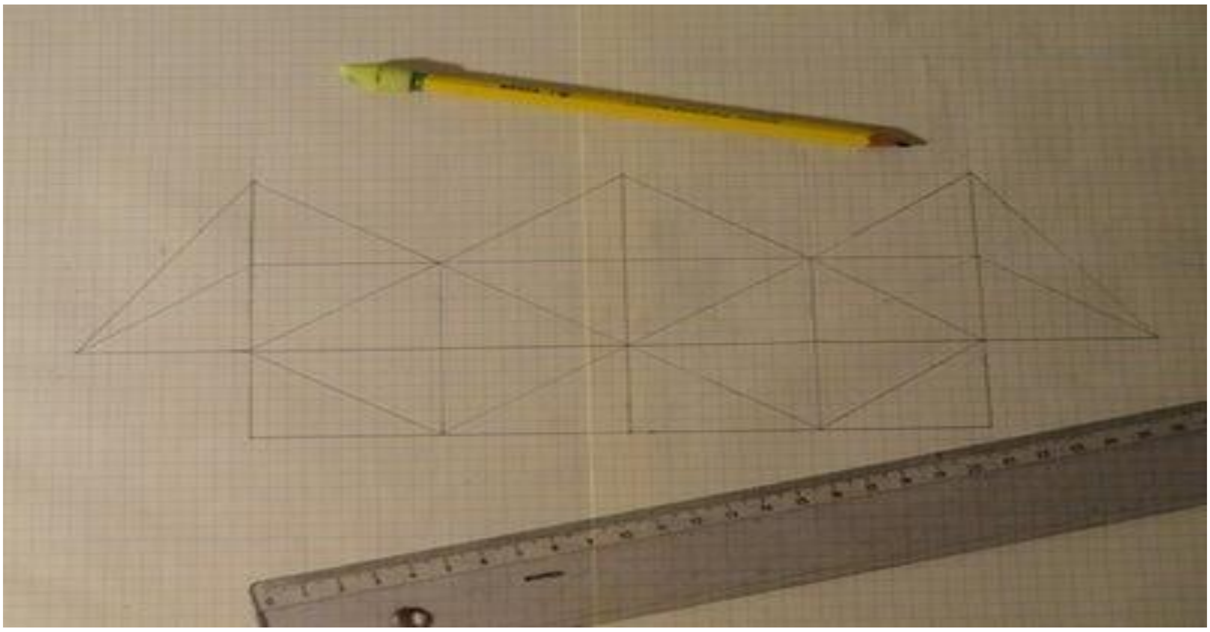
Step 1: Gathering Your Materials

You will not need a lot of materials and they are all super cheap and easy to find.

- Graph Paper
- Markers
- Pencil
- Bottle Cap
- Multi-Purpose Glue
- Plastic Wrap
- Toothpicks

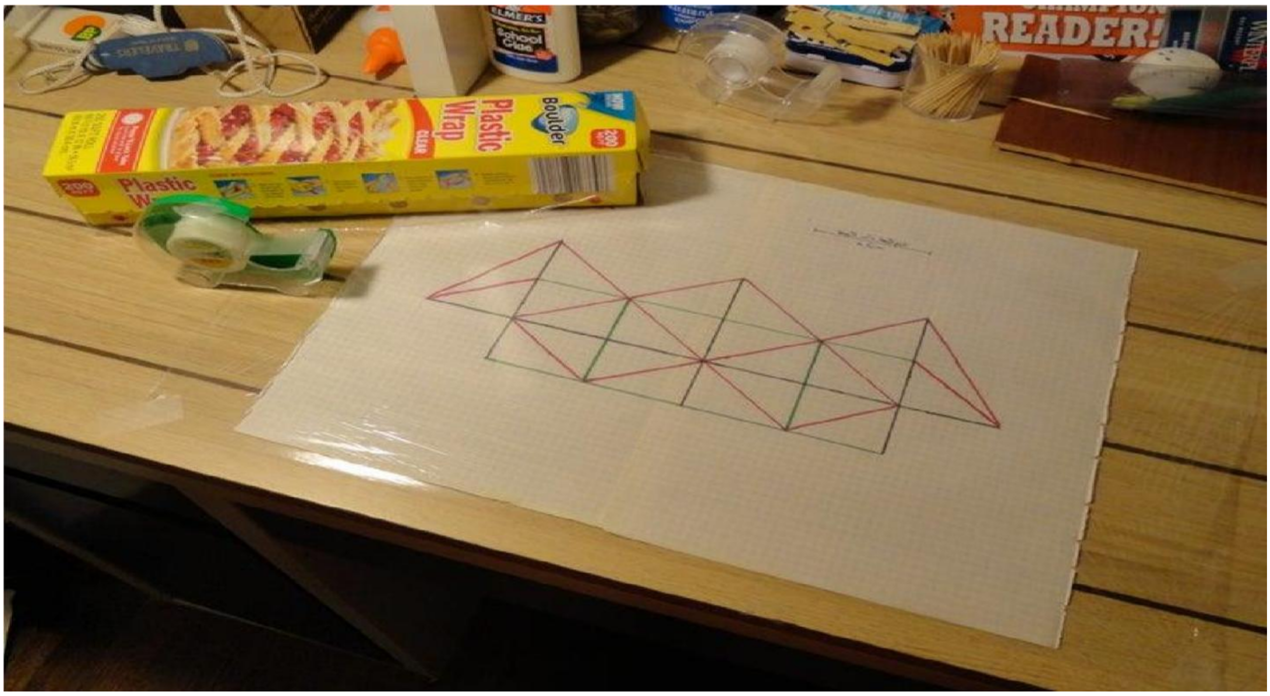
Step 2: Drawing the Design





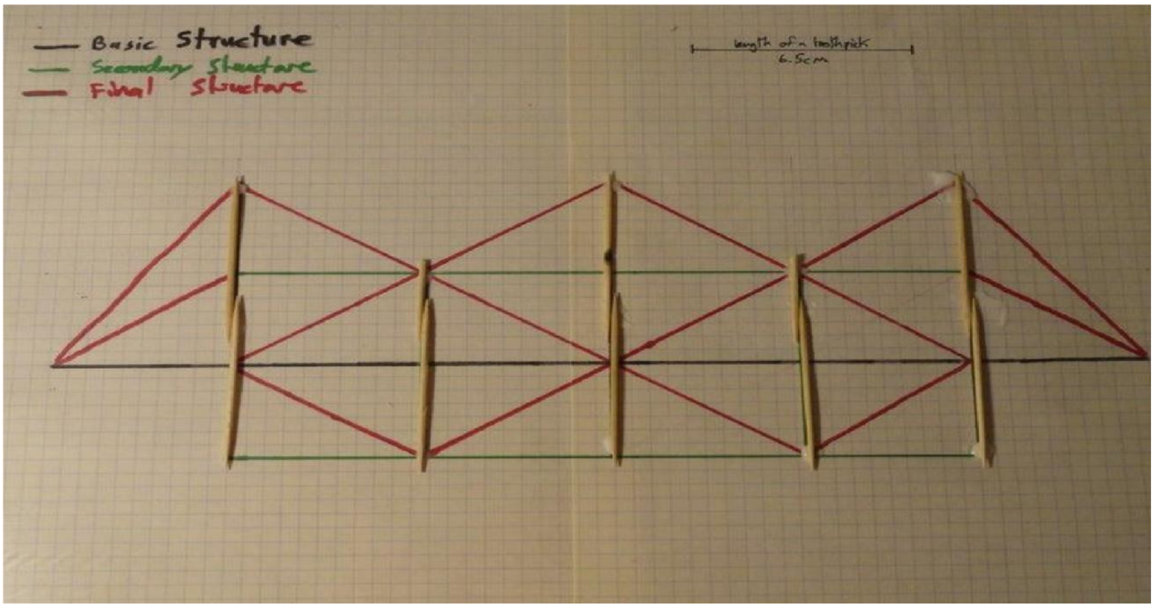
1. Tape two pieces of graph paper together.
2. Sketch the bridge on to the paper with the pencil.
3. With different coloured makers draw the Bridge showing the order you will glue the toothpicks.

Step 3: Placing the Plastic Wrap



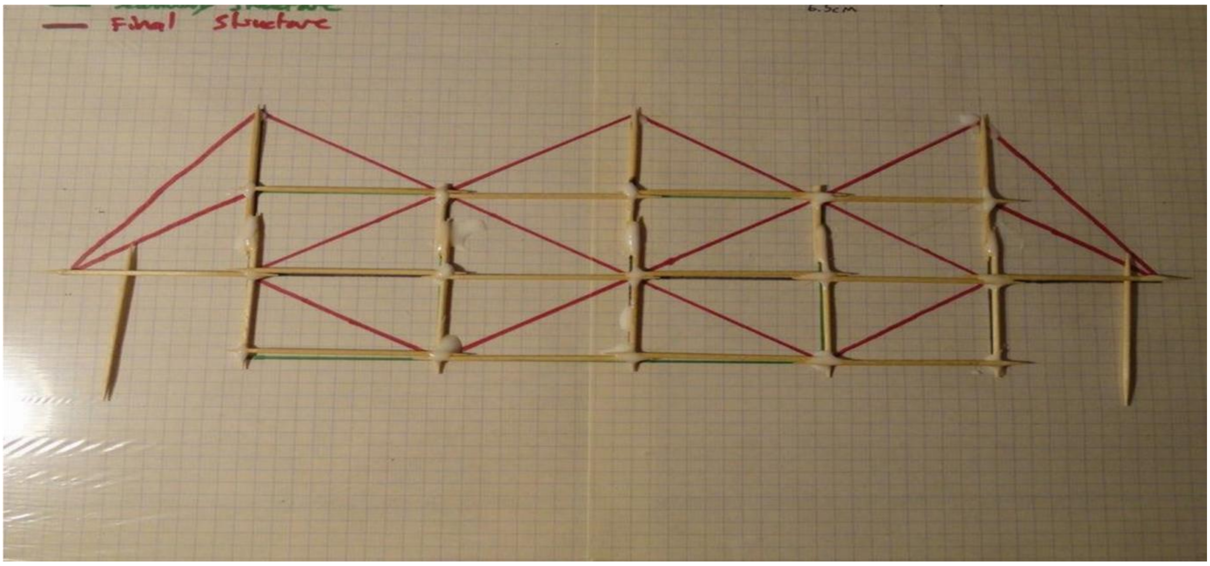
1. Put the drawing of the bridge on a flat smooth surface.
2. Then lay the plastic wrap over it and tape it down you want the plastic wrap to be perfectly smooth.

Step 4: Making the Basic Structure



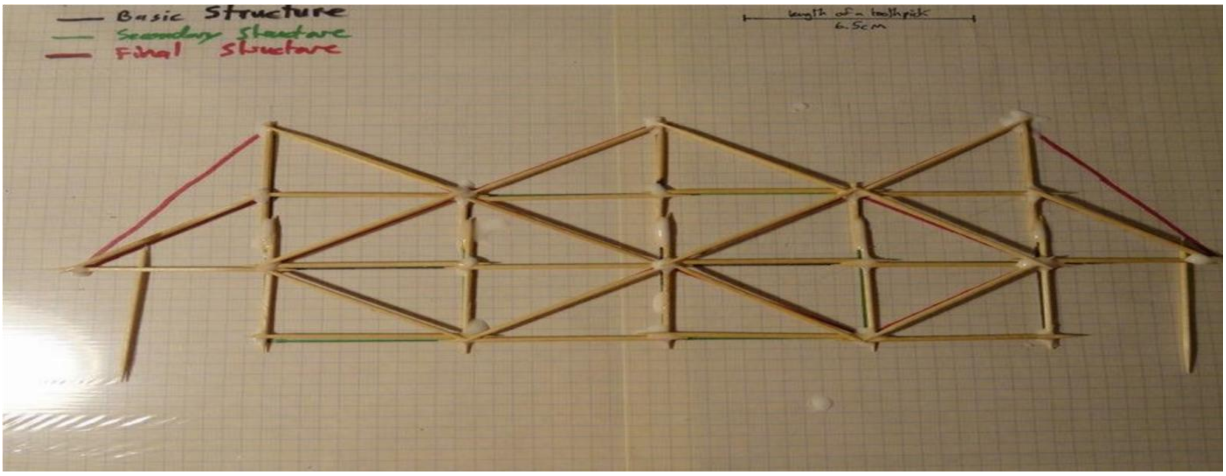
1. Cut the toothpicks to the right size.
2. Lay down all the toothpicks that are supposed to be vertical.
3. Put glue on the toothpicks.
4. Put the horizontal toothpicks that are on the basic structure.
5. Glue everything together.

Step 5: Making the Secondary Structure



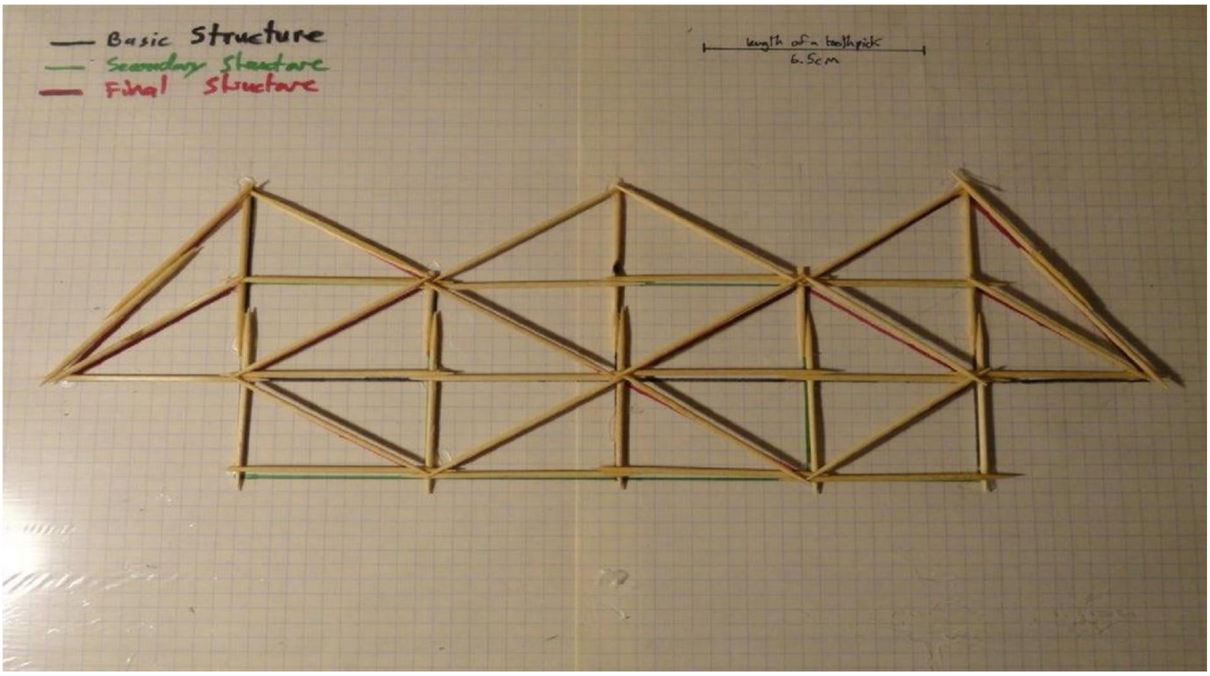
1. Place all the toothpicks on the secondary structure
2. Glue them all down.

Step 6: Making the Final Structure



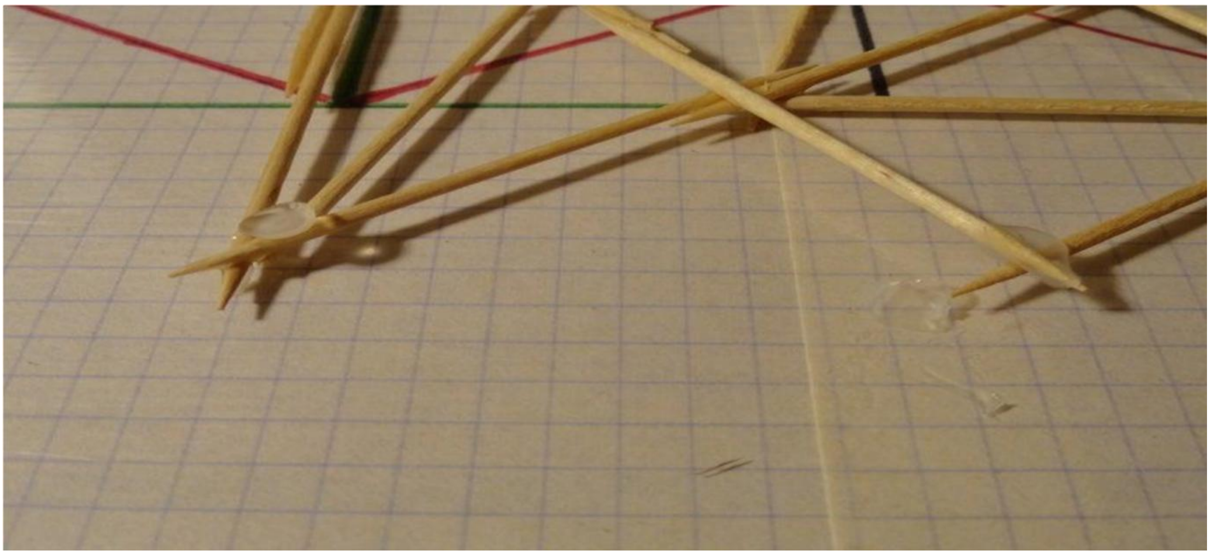
1. Place all the toothpicks on the Final structure except for the long piece on the sides.
2. Glue them all down.
3. Make the two long pieces aside.
4. Let the bridge dry for at least 12 hours.

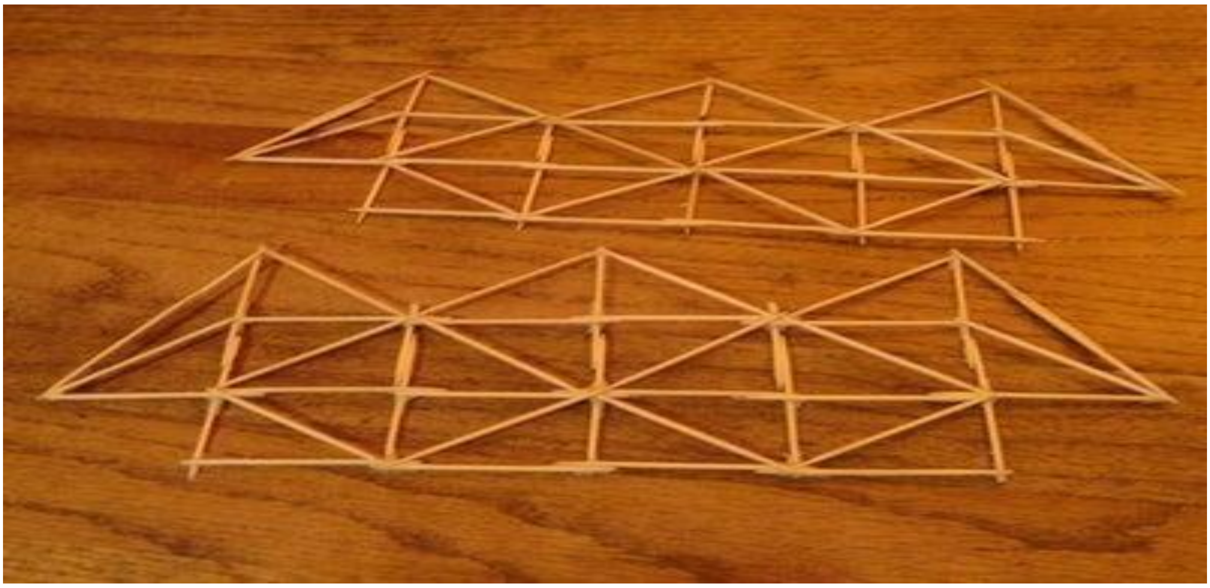
Step 7: Placing the Side Pieces



1. Place them on the side of the bridge.
2. Glue them down.
3. Let it dry for another 12 hours.

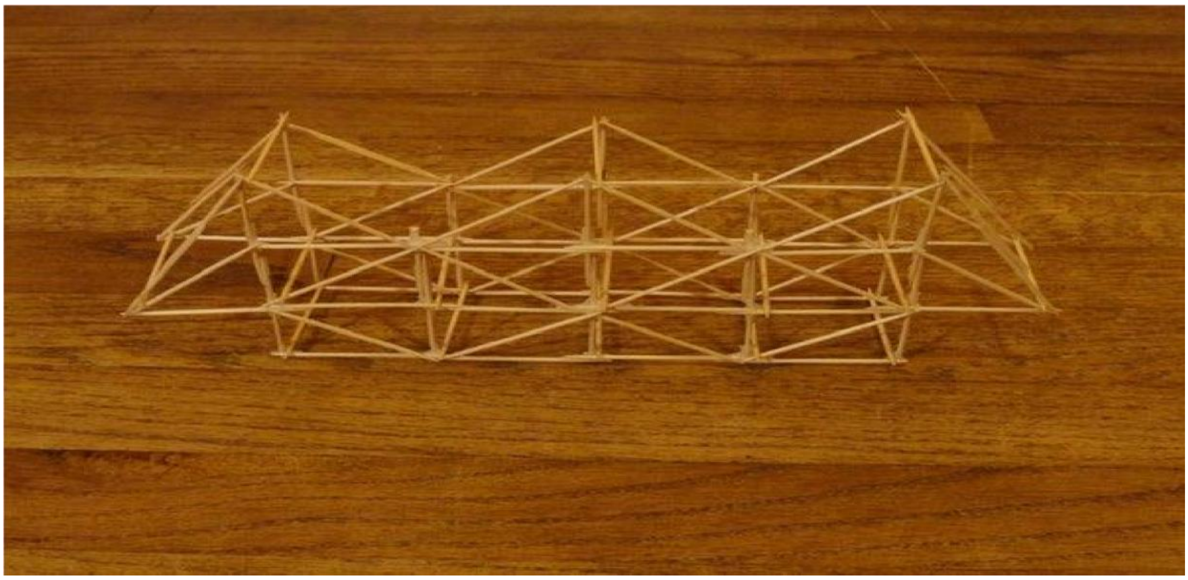
Step 8: Taking the Bridge Off From the Plastic Wrap

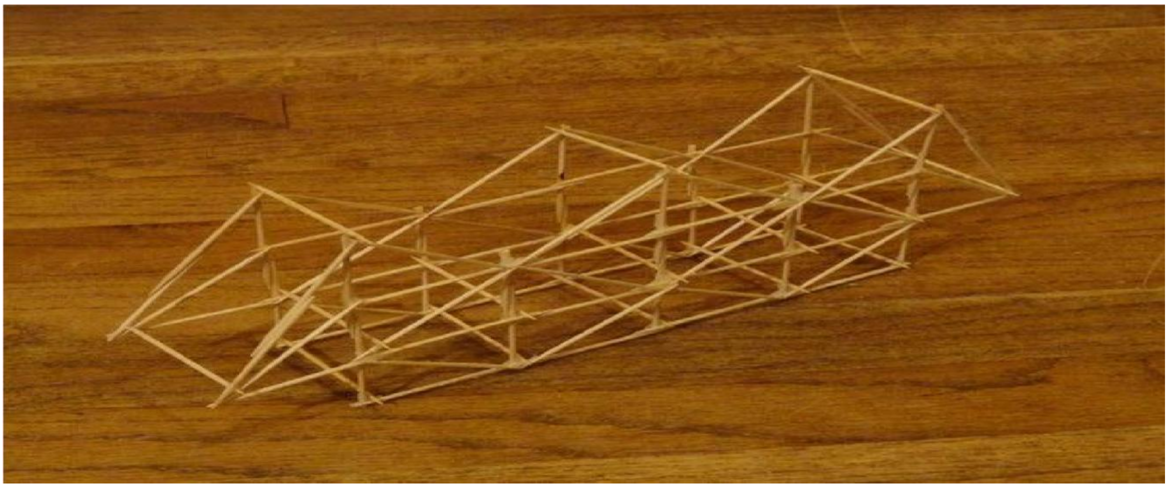




1. Take the bridge off from the plastic wrap carefully.
2. Take of all the pieces of ex’s glue from the bridge.
3. Repeat steps 3-8 to make the other side of the bridge.

Step 9: Assembling the Bridge

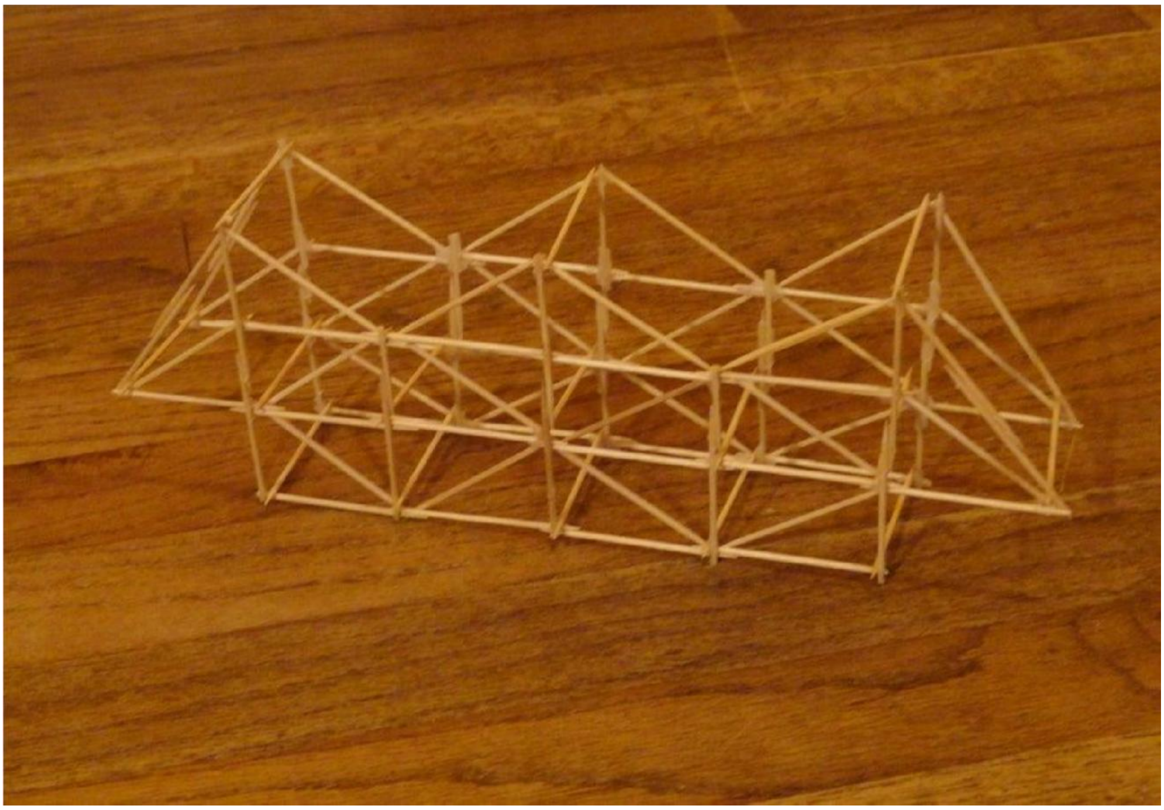




1. Place the two bridge pieces parallel to each other one toothpick length from each other.

2. Place toothpicks between the two parts of the bridge.
3. Glue the toothpicks.
4. Let the Bridge dry for 12 hours.
5. Bridge Finished.

Step 10: Conclusion



Experiment-4

AIM

Prepared a marble maze (mock up model)

Description

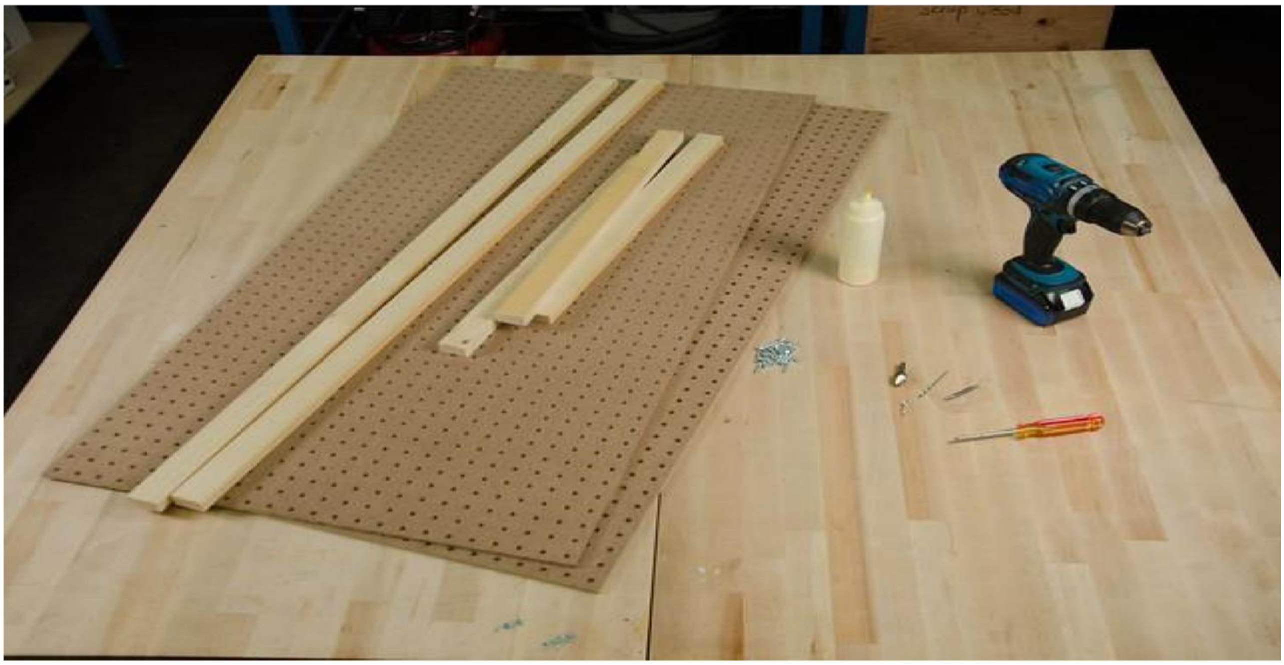
Make a Marble Machines Board

A Marble Machine is a creative ball -run contraption, made from familiar materials, designed to send a rolling marble through tubes and funnels, across tracks and bumpers, and into a catch at the end.

In this activity guide, you'll learn how to build a pegboard panel that will allow you to experiment with a variety of household materials to build elaborate marble runs. Once you have the basic board built, you will get hours of play out of it, and you will be able to try more ambitious ideas.

Step 1: Materials

This project requires some materials, easily obtainable from your local hardware store. You will need:



- 2 sheets of **pegboard** - 2 feet x 4 feet
- 2 lengths of wood - 1 in x 2 in x 4 feet (**vertical spacers**)
- 3 lengths of wood - 1 in x 2 in x 22.5 in (**horizontal spacers**)
- wood glue (optional but recommended)
- wood screws
- power drill
- drill bit and driver bit
- countersink bit (optional)

- 1/4 in **dowels** (you will use these pegs for alignment)
- a friend (recommended)

Step 2: Align Pegboard and Vertical Spacers

Set one of the pegboard panels on top of the two long vertical spacers. Do this with both spacers at once to keep the panel level. Make sure edges and sides of the spacers are flush with the panel. Just running your fingers along the edge will do the trick.

Tip: Long pieces of wood are often slighted curved. This is not a problem, as they can be straightened when fastening them to the pegboard, but make sure they are flush on one end to start, and straighten them as you work your way along.

Step 3: Pre-drill Pegboard and Vertical Spacer

1. Once your spacers are aligned and flush, you'll pre-drill all your screw holes to avoid splitting the wood with screws. Start at one end, and drill a small hole where the first screw will go.

Tip: If you want your board to look extra slick and have all the screws be flush, you can use a countersink bit to make room for the screw heads. Now is the time to do so.

2. Put the first screw in to tack one end in place while you pre-drill the rest. If you plan on using glue, don't put the rest of the screws in yet.

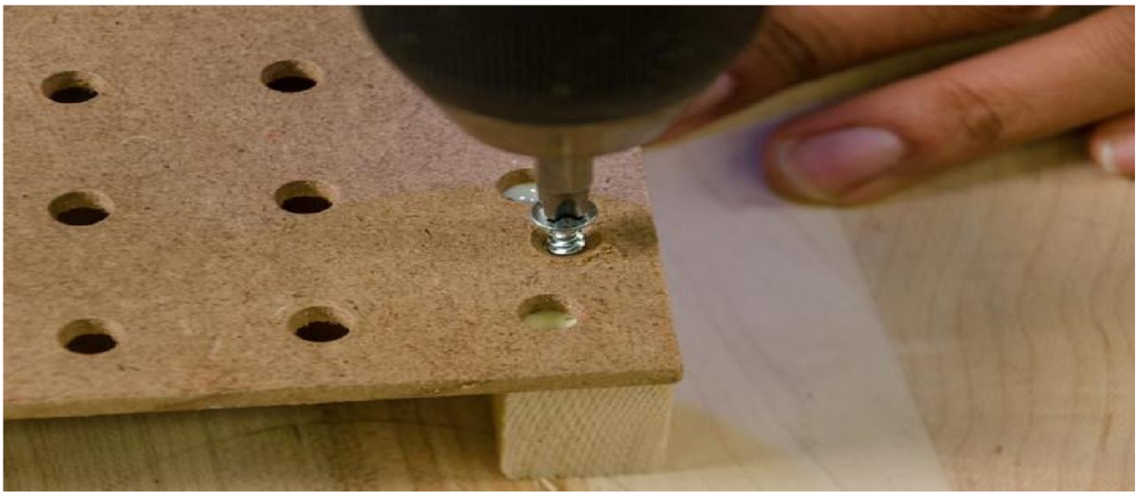
If you're skipping the glue, feel free to add screws as you go along.

3. Pre-drill every 6 inches or so down the length of the board. If you notice the spacer is warped or bent, pull, or push it flush with the edge of the board as you go. This will help keep things aligned when you permanently attach the board to the spacers.

Step 4: Attach Spacers

1. If you've decided to use wood glue, lift the board off the spacers, and apply glue all the way down the one you pre-drilled. Use a wavy pattern for better coverage and style. If you've decided not to use glue, you can leave the board in place.

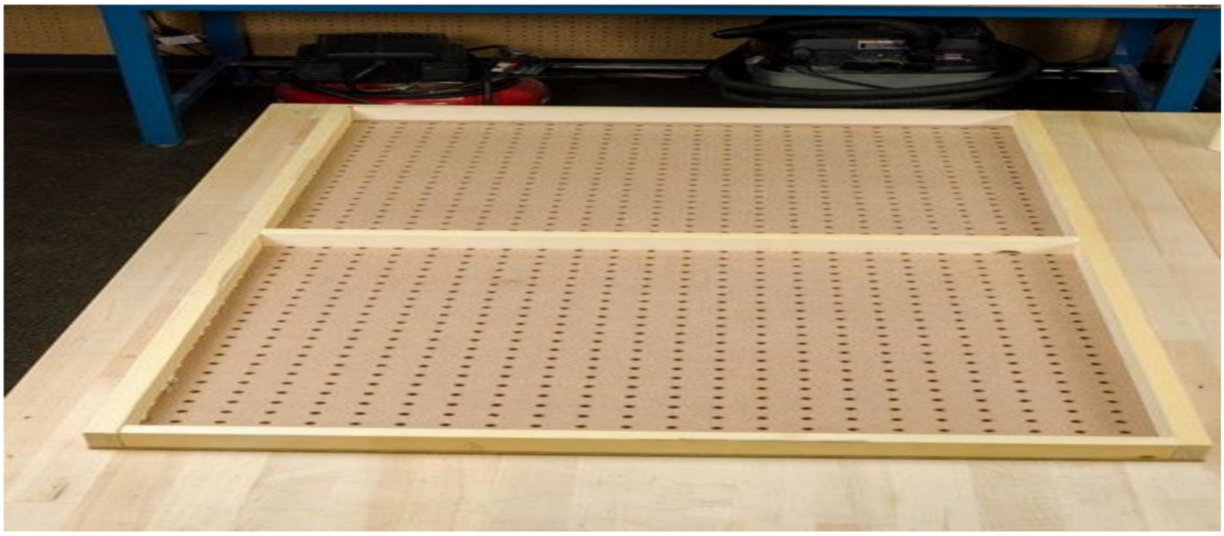
2. Put the board on top of the spacers again, and drive screws in the holes you pre-drilled. The screws should find the pre-drilled holes and line everything up nicely!



It's normal for some glue to squeeze out; use a damp cloth to wipe it up (your finger will also work in a pinch).

3. Repeat the whole process with the other vertical spacer.

Step 5: Place the Horizontal Spacers



1. Next, you will add horizontal spacers to your board. These provide structural integrity and prevent the pegboard panels from bowing inward.

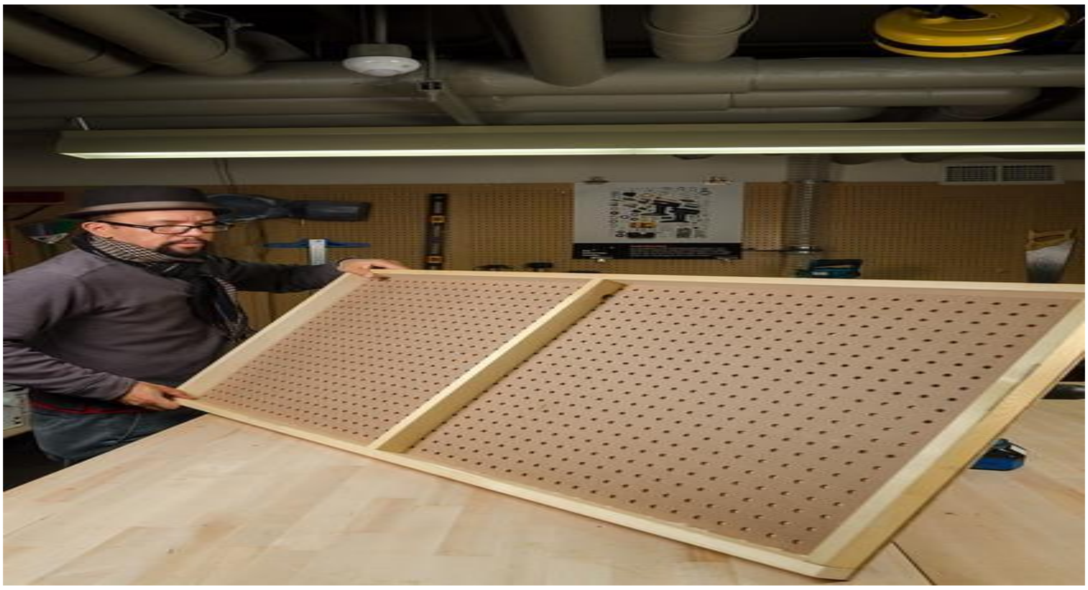
2. Flip your board and put a horizontal spacer on each end and one in the middle. Take care to place the middle spacer so it doesn't block any of the holes.

3. If the fit is tight, the spacers will stay in place. If not, use masking tape to keep them in place.

4. Flip the board over again and pre-drill, countersink (optional), and screw the horizontal spacers in place.

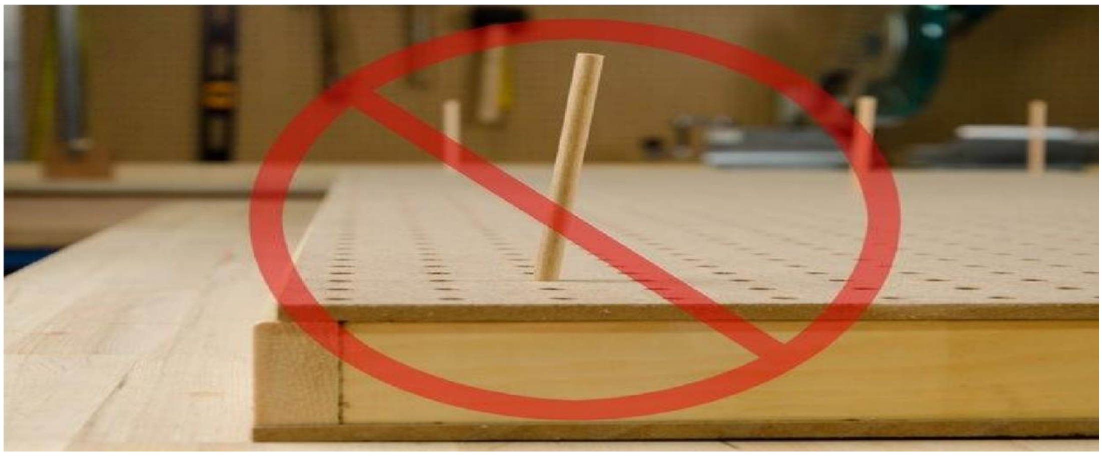
Step 6: Add the Second Pegboard Panel

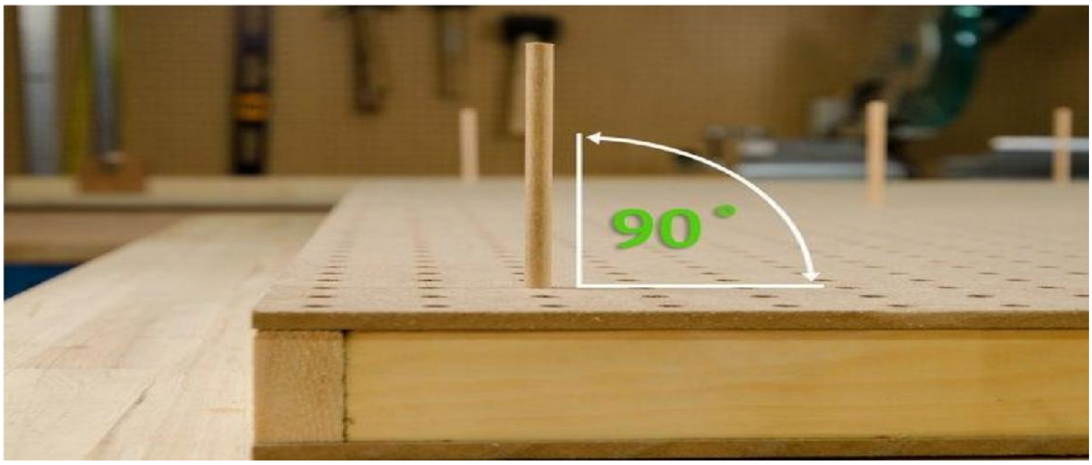
1. Flip your board over again.



2. Place the second pegboard panel on top.

3. Use at least three dowels to align the holes in the top panel with the one on the bottom. It is important to make sure the dowels will stick straight out when you're building your marble run.





4. Stick dowels through both sets of holes, and visually check that they are vertical. Often, fixing a dowel in one spot will make others lean; keep adjusting until they are all as close to vertical as you can get. It doesn't have to be perfect, close is good enough.

5. Now you're ready to pre-drill and countersink (optional) the whole panel. Don't forget to pre-drill the horizontal spacers too, including the one in the middle, which is now hidden.

6. If you're not using glue, you can screw the panel in now. If using glue, lift it off, apply glue, and replace it before screwing it in.

Tip: Sometimes it's helpful to use dowels again to confirm the panels are aligned before screwing it together.

Step 7: Your Board Is Nearly Done.

Take a moment to be proud of your board. The hardest part is done. When you're ready, move on to the next step: make feet so your board can stand on its own.

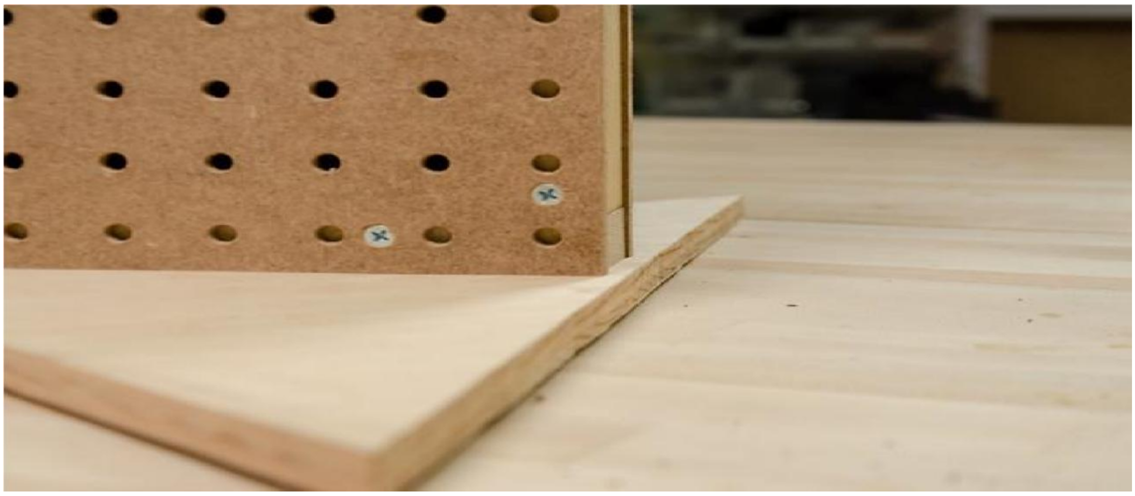
Step 8: On Your Feet!

1. Find or cut a piece of plywood that measures about 15 inches square. Draw a line from corner to corner and cut the square into two equal triangles (feet).

2. Locate the centre of the long side of each triangle and mark it. If you draw a line with a ruler from this mark to the point on the other side you'll have a nice centreline.

3. Decide which side of your marble board you want to have as the bottom. Use the centreline on one of your feet with to centre it on the edge of your board.

4. Instead of making the foot flush with the bottom, make it stand out, or "proud," by 1/4 inch. This will make your board much more steady, particularly if the floor isn't perfectly flat.



5. You should pre-drill, and countersink (optional), and screw in one of the feet. Then flip the board and repeat with the other side.

Tip: This is a great time to have a friend helping because it's tricky to keep the board steady when it's standing on end and you are trying to drive screws into it.

6. A quicker and easier way to make feet is to use a straight piece of wood about 18 or 20 inches long, and screw that in place. It won't be as durable as the triangle feet but it will work. Remember to keep it "proud" of the bottom by 1/4 inch.

Step 9: Hooray! a Completed Marble Board Masterpiece

Experiment-5

AIM

Build a wind power car (mock up model)

Description

Wind-Powered Car

This project will show you how to create a wind-powered car with a laser cut cardboard body, 3D printed wheels and a paper sail. **Supplies**

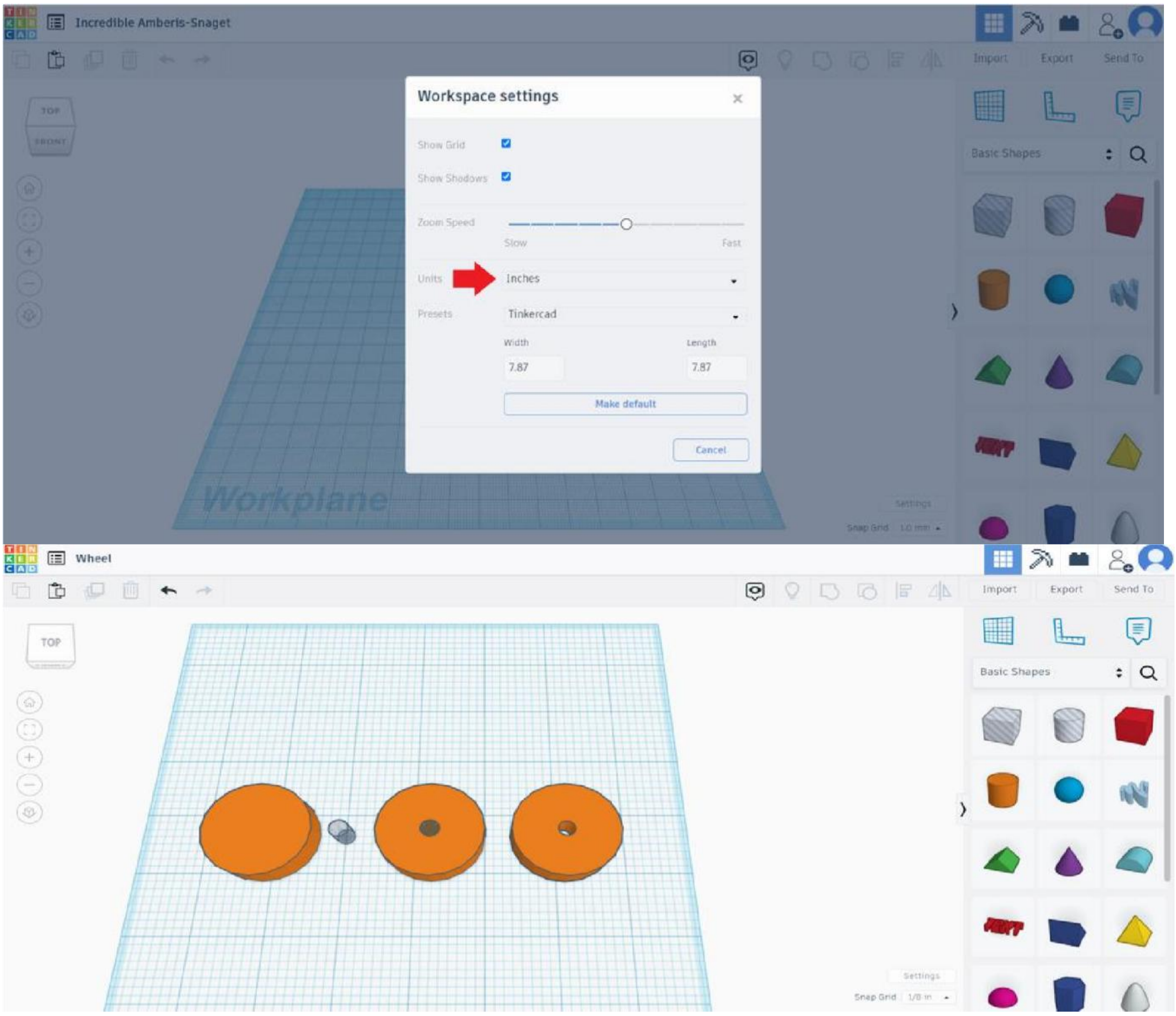
- Cardboard
- Wooden Dowels (I used 1/4 x 12 inch)
- Cardstock
- Glue
- Paint and Paint Brushes
- Coloured Pencils/Markers
- Tinker cad
- Glow forge (or another laser cutter)

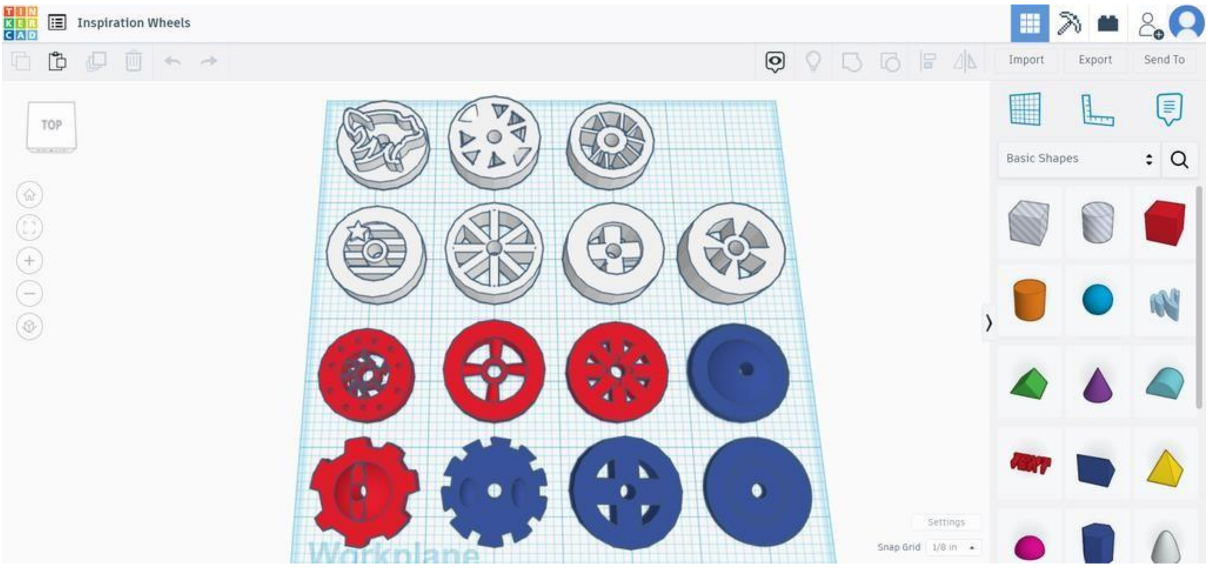
- 3D Printer and Filament
- Cricut (or scissors)

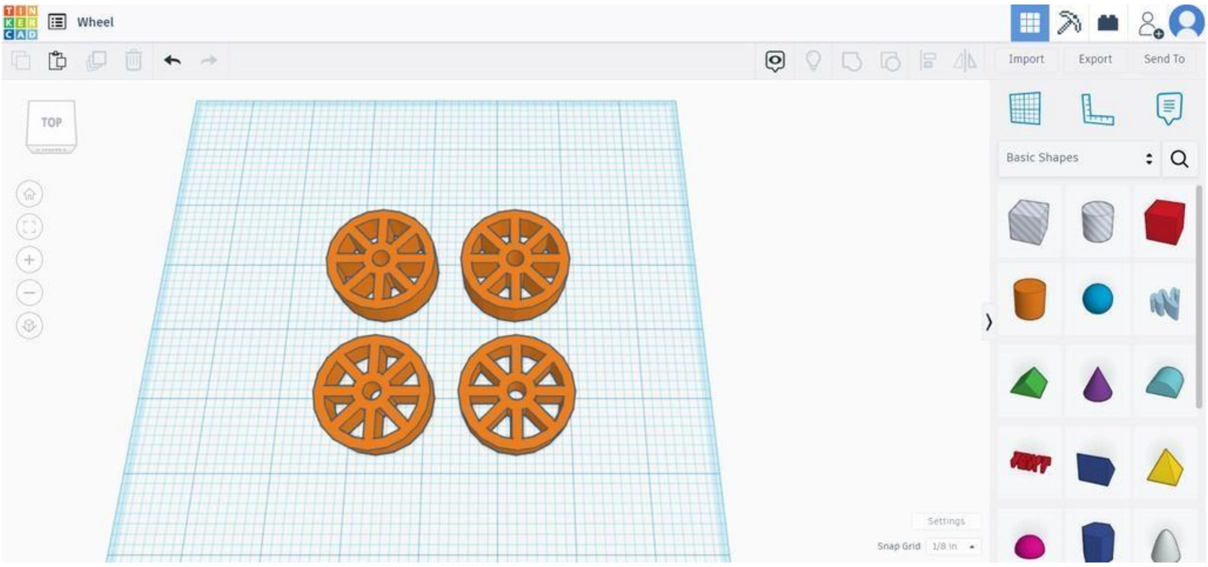
Step 1: Plan Your Design

Start off by sketching what you would like the body, wheels and sail of your car to look like. I asked my students to create at least 2 designs for each part of their car. Then they picked their favorite ones.

Step 2: Design Your Wheels



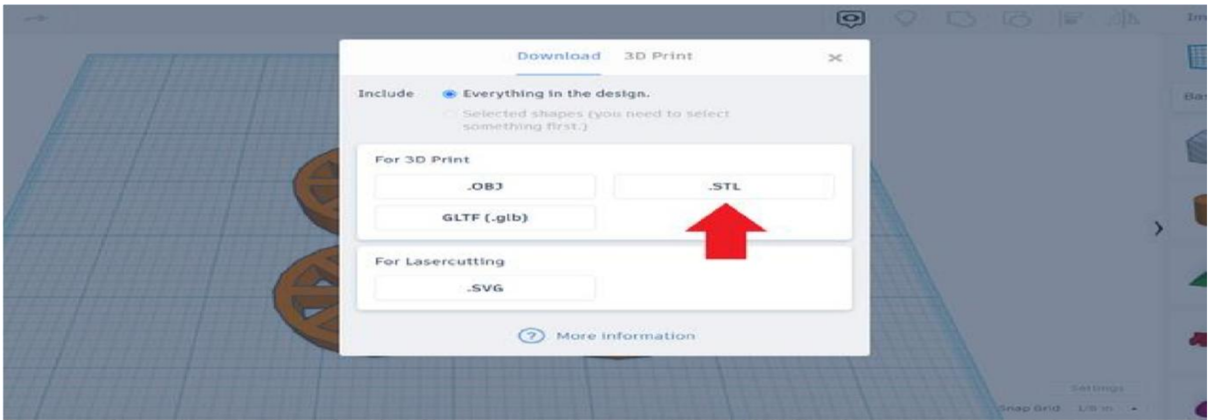


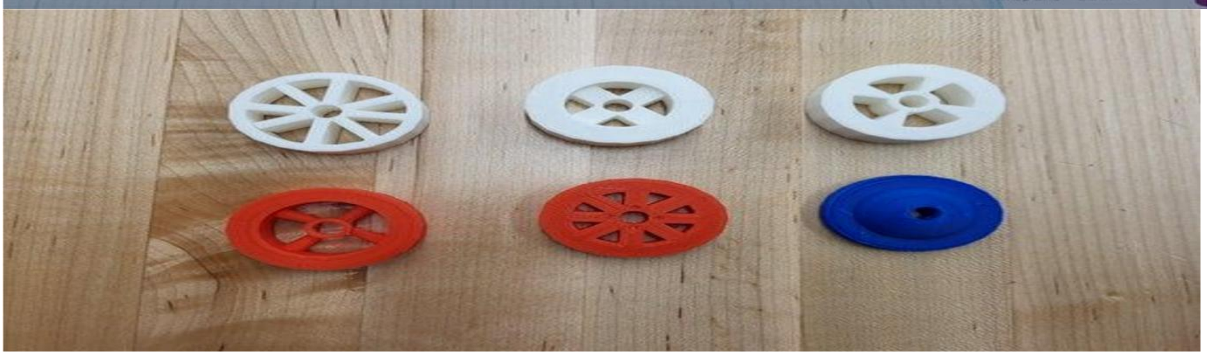


1. Create a new 3D design in Tinker cad.
2. If your dowel is measured in inches like mine, go into the Settings (in the bottom right corner) and switch the units to inches.
3. Drag a cylinder onto your work plane and resize it to the dimensions you would like your wheels to be. I gave my students a max diameter of 2 inches and a max height of 1/2 inch.
4. Drag a cylinder hole out, resize it to 1/4 by 1/4 inch (or whatever size your dowel is) and adjust the height to match your cylinder.

5. Select both shapes and use the align tool to align the hole to the center of your wheel.
6. Group both shapes. This will cut a hole out of your wheel for the dowel.
7. If you'd like, you can use additional shapes to get more creative with your wheel design. Check out my students' awesome creations in the images above if you need inspiration!
8. Once you have your wheel exactly how you would like it, duplicate it 3 times. Space your wheels out on the work plane.

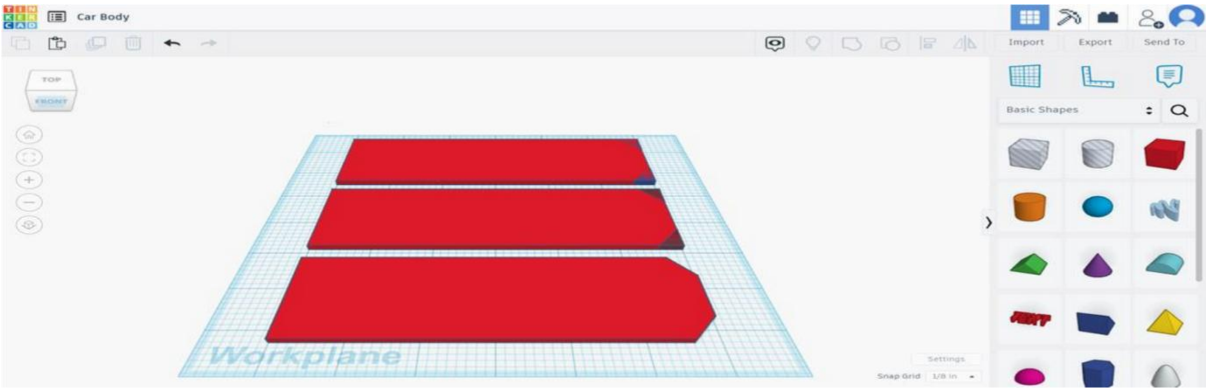
Step 3: 3D Print Your Wheels

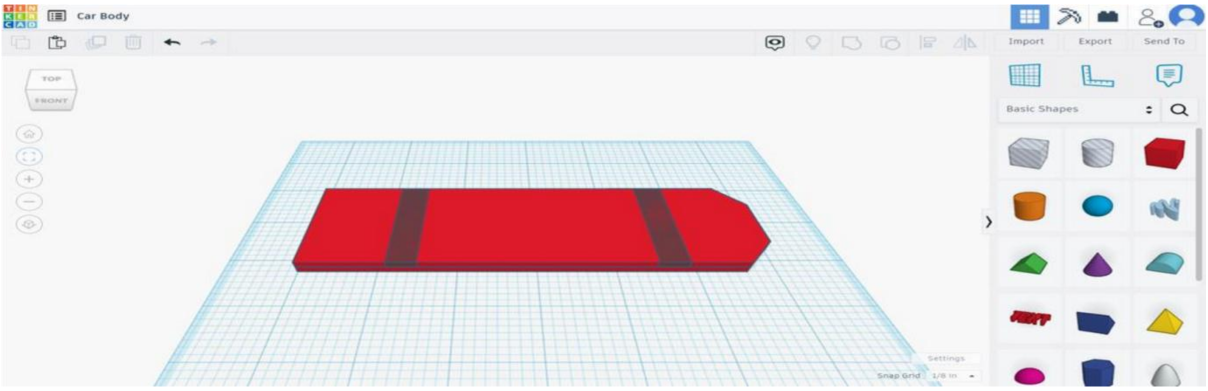


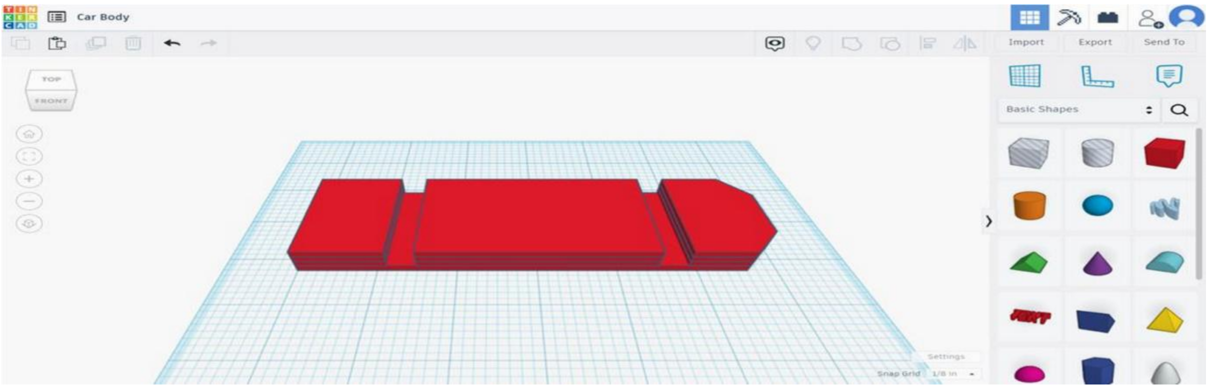


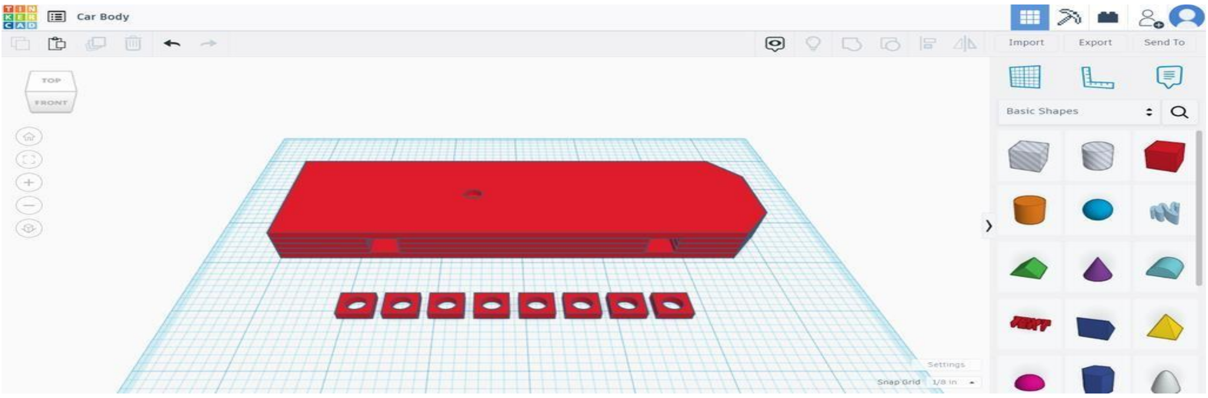
Export your design as an .STL file and print it on your 3D printer.

Step 4: Design Your Car Body









The body of the car is made out stacked layers of cardboard. In this next step, you'll be designing your layer by layer. To do so:

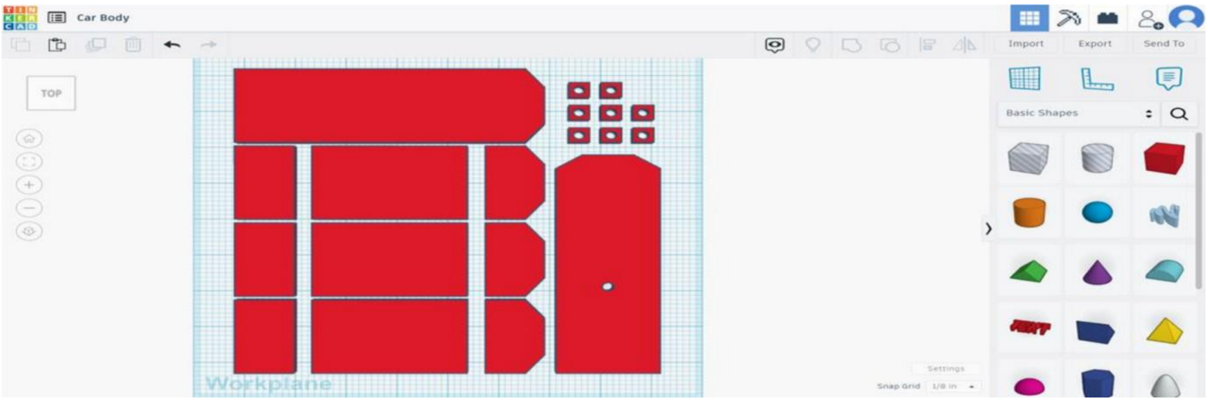
1. Create a new 3D design in Tinker cad.
2. Go into Settings again and switch the units to inches. If you're using cardboard boxes (like Amazon boxes), the cardboard is about 1/8 inch thick.
3. Create the base layer of your car using a single shape or multiple shapes grouped together. I gave my students a max length of 8 inches and a max width of 3 inches minus the height of 2 of their wheels.
4. Adjust the height of your base layer to 1/8 inch.
5. Duplicate your base layer and raise it 1/8 inch so it is directly above your first layer.

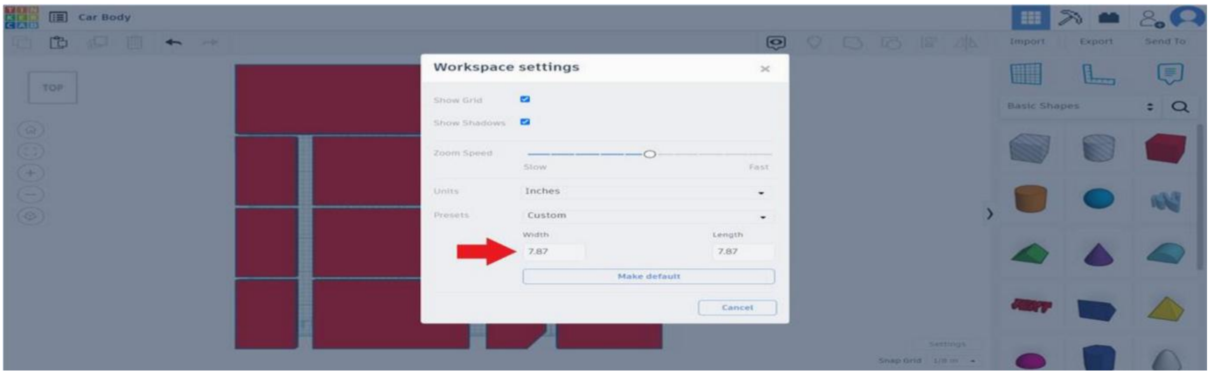
6. Drag a box hole out. Resize it so the height is 1/8 inch, the length is 3/8 inch and the width is the same as your second layer.
7. Raise the box hole 1/8 inch and position it in your second layer where you would like one of your axels to go.
8. Duplicate the box hole and nudge it over (using the arrow keys) to where you would like your other axel to go.
9. Select your second layer and both holes. Group them together.
10. Duplicate your second layer and raise it 1/8 inch to make your third layer.
11. Duplicate again to add a fourth layer. (Raise it 1/8 inch if it doesn't do so automatically)
12. Select your base layer, duplicate it and raise it 1/2 inch so it is directly above your fourth layer.
13. Drag a cylinder hole out. Resize it to 1/4 by 1/4 inch and a height of 1/8 inch.
14. Raise the cylinder hole 1/2 inch and position it where you would like the mast for your sail to go.
15. Select your fifth layer and the hole and use the align tool to align the hole to the middle of your fifth layer. Then group both shapes.
16. If you'd like, you can add additional layers to your car body for a more creative design. You can use different sizes and shapes.

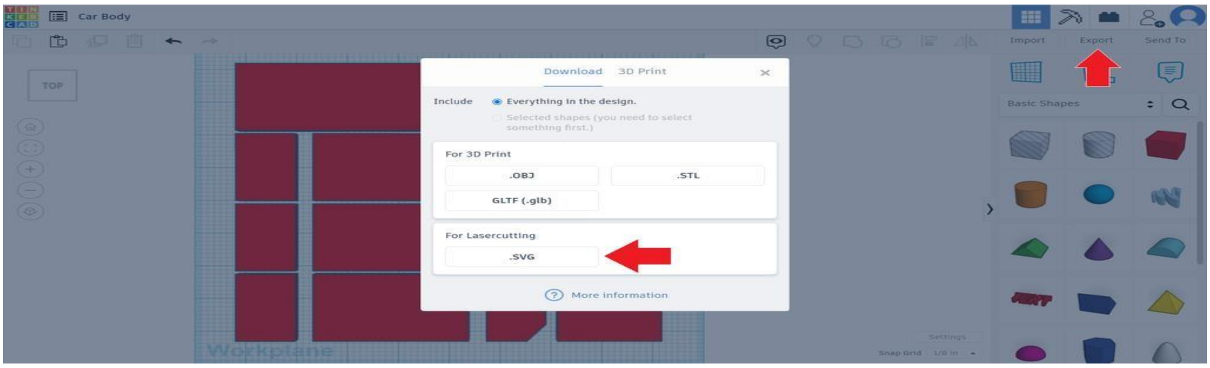
You also need to make holders for your axels so they won't wobble. These holders will go in the holes you cut out in your second, third and fourth layers. To make them:

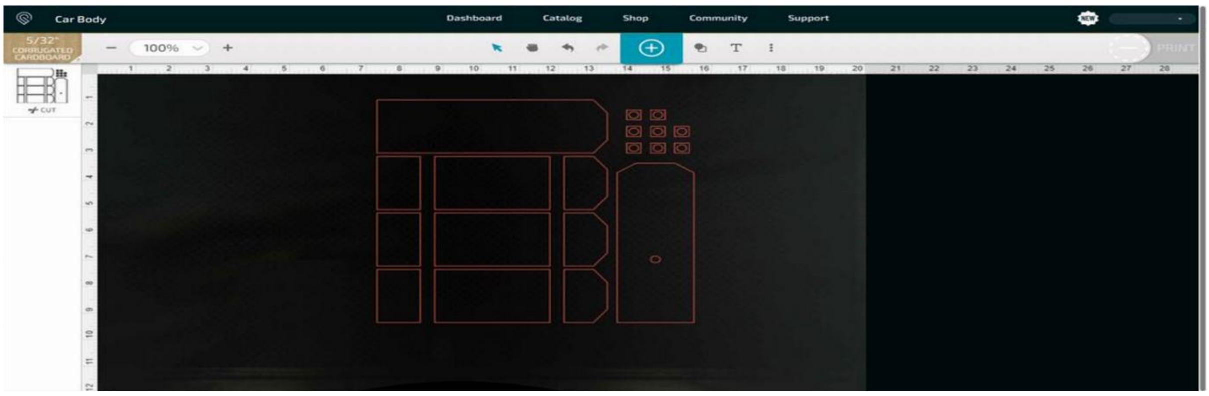
1. Drag a box onto your workplane. Resize it to 3/8 by 3/8 inch and a height of 1/8 inch.
2. Drag a cylinder hole out. Resize it to 1/4 by 1/4 inch (or whatever size your dowel is) and a height of 1/8 inch.
3. Select both shapes and use the align tool to align the hole to the center of the box. Then group both shapes.
4. Duplicate the holder 7 times and space them out on the workplane.

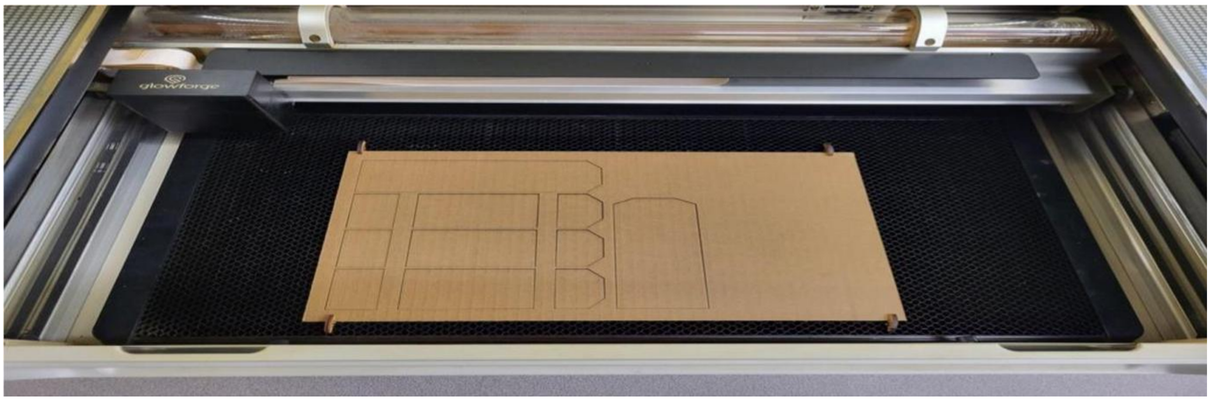
Step 5: Cut Your Car Body











To prepare your design for laser cutting:

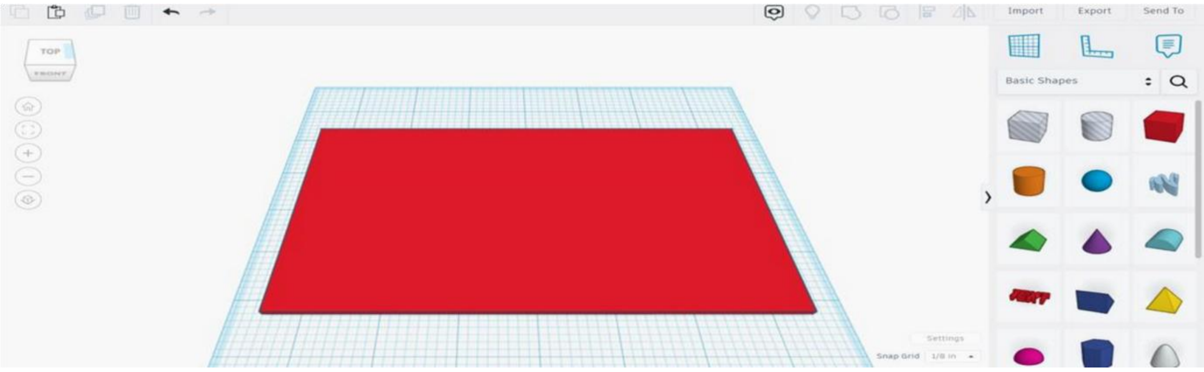
1. Click on each layer of your car body and hit the D key on your keyboard. This will drop each layer down onto the workplane.

2. Space all your pieces out on the workplane. If you run out of room, go into Settings and make the dimensions of your workplane larger.

3. Export your design as an .SVG file.

Upload the .SVG into the app for your Glow forge or other laser cutter and cut your pieces out. If you would like, you can paint your pieces once cut!

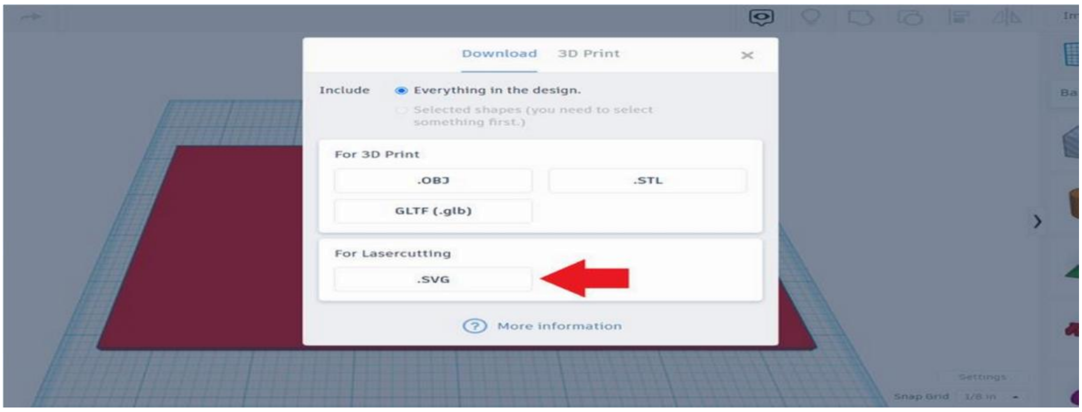
Step 6: Design the Sail

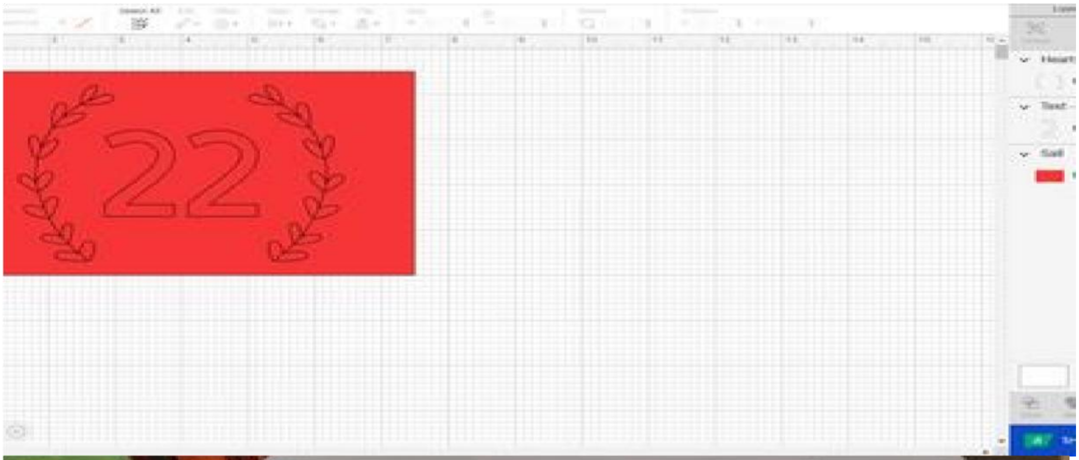


To make your sail:

1. Create a new 3D design in Tinker cad.
2. Once again go into Settings and change the units to inches.
3. Use a single shape or multiple shapes grouped together to create the shape of your sail. I gave my students max dimensions of 10 by 7.5 inches so that the Cricut would be able to cut their sail out of a piece of letter size cardstock.
4. The height doesn't really matter for this step but I had my students adjust theirs to 1/8 inch so it looked more like paper.

Step 7: Cut the Sail





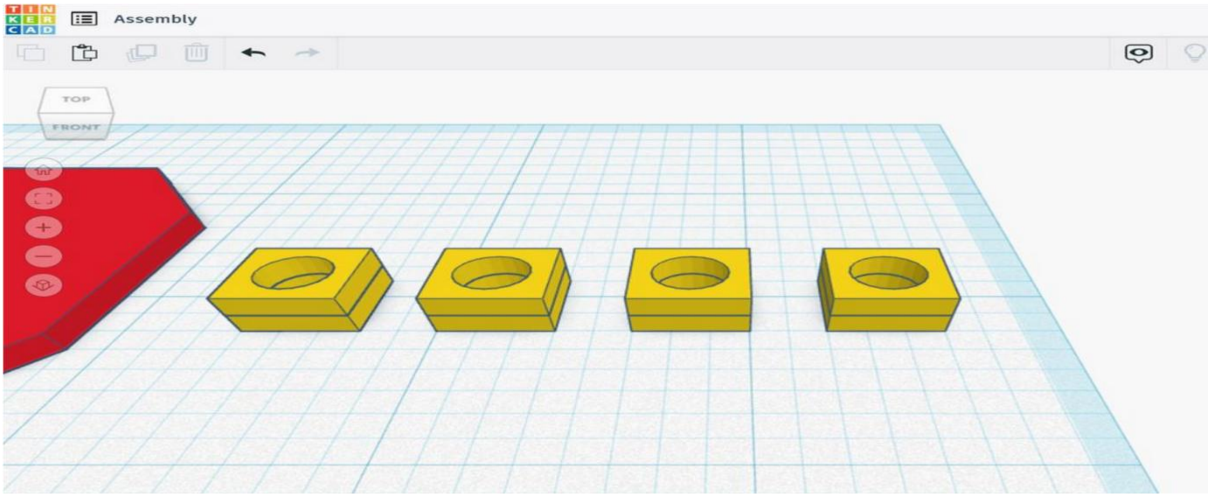
Export your design as an .SVG file. Upload it into the app for your Cricut or other paper cutter.

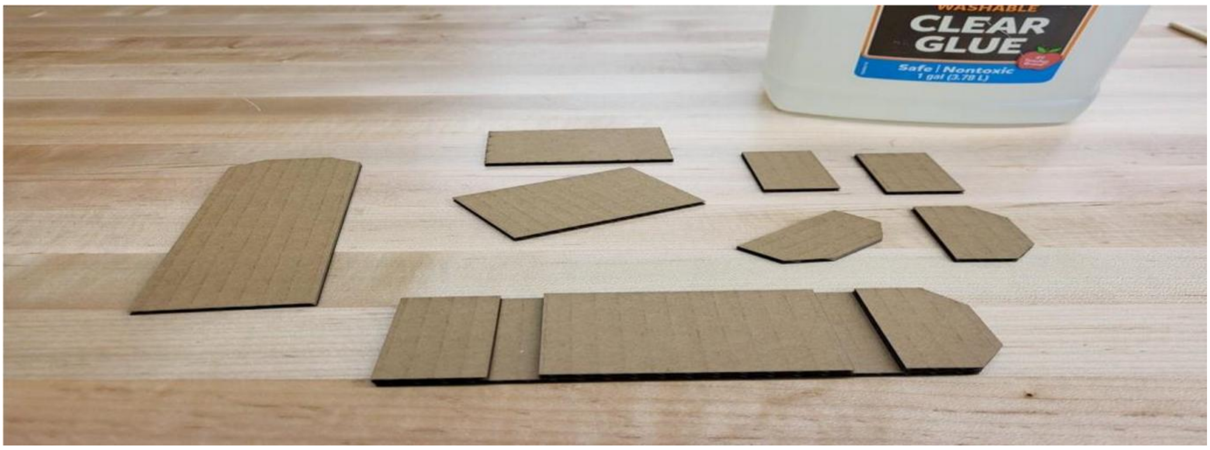
Before cutting, I let my students add a number and other images to their sail. We switched these layers from "Basic Cut" to "Pen" in order to have the Cricut draw these items on the sail.

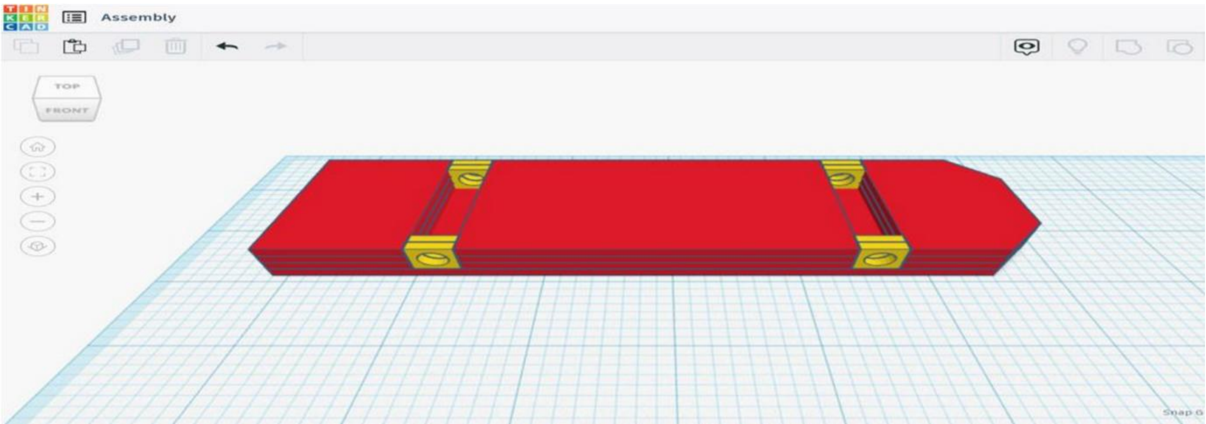
Cut out your sail. If you don't have a Cricut or other paper cutter, you can always cut out your sail with a good old-fashioned pair of scissors! If you would like, you can also color your sail with colored pencils or markers.

Step 8: Cut Your Axels and Mast

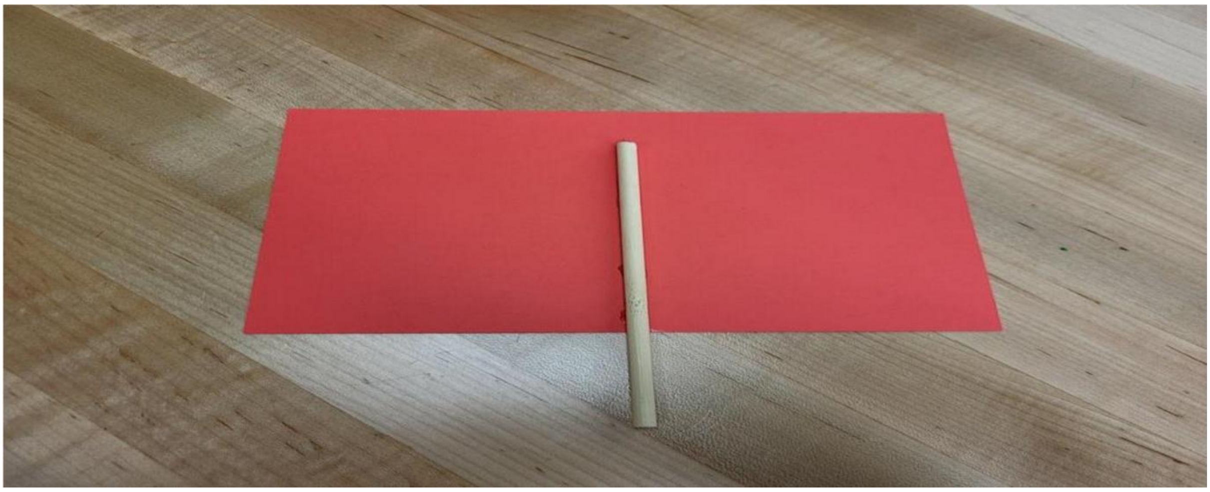
Cut your dowel into two 3-inch axels and a 6-inch mast.







Step 9: Assemble Your Car



To assemble your car:

1. Glue one of your axel holders on top of another. Repeat these 3 more times so that you have 4 doubled up axel holders. (I forgot to take picture of this step - see the first Tinker cad image above instead)

2. Glue your second, third and fourth layers on top of the base layer of your car body. (TIP: Use your axel holders as spacers)
3. Glue the axel holders on their sides in the holes. (See second Tinker cad image above)
4. Glue on your fifth layer on top, along with any additional layers.
5. Glue your sail to your mast. (I used hot glue)
6. Glue your mast into the mast hole in your fifth layer. (I used hot glue)
7. Thread your axels through the axel holders and attach your wheels to both sides.

Step 10: Race!

Put your car in front of a fan, turn it on and see how far it goes!

Experiment-6

AIM

Make a hydraulic elevator (mock up models)

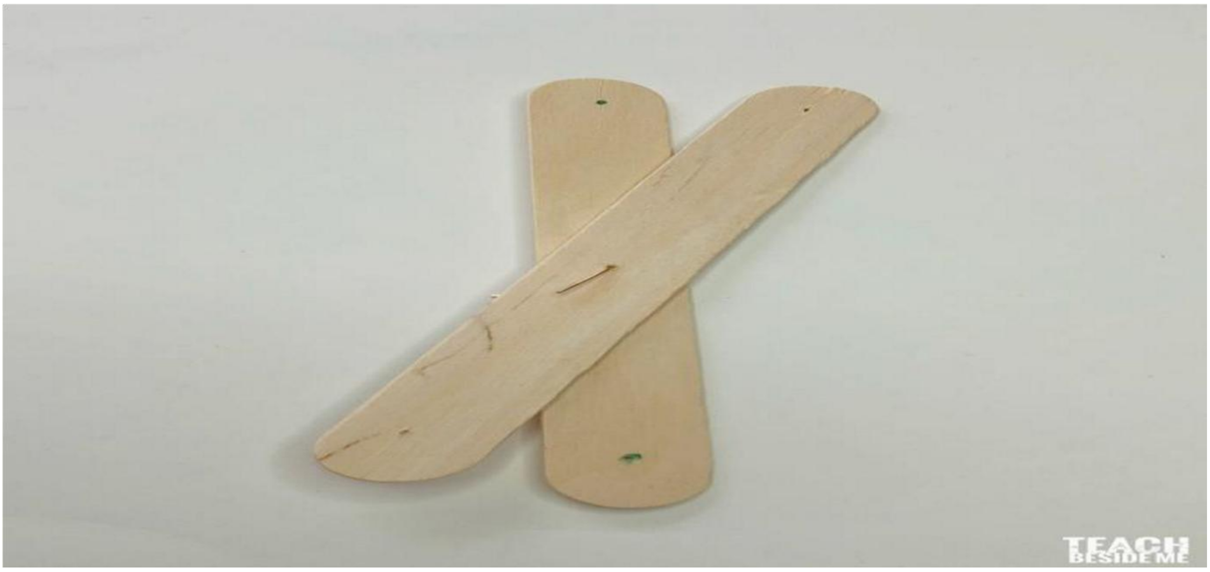
Description

SUPPLIES FOR THE HYDRAULIC ELEVATOR:

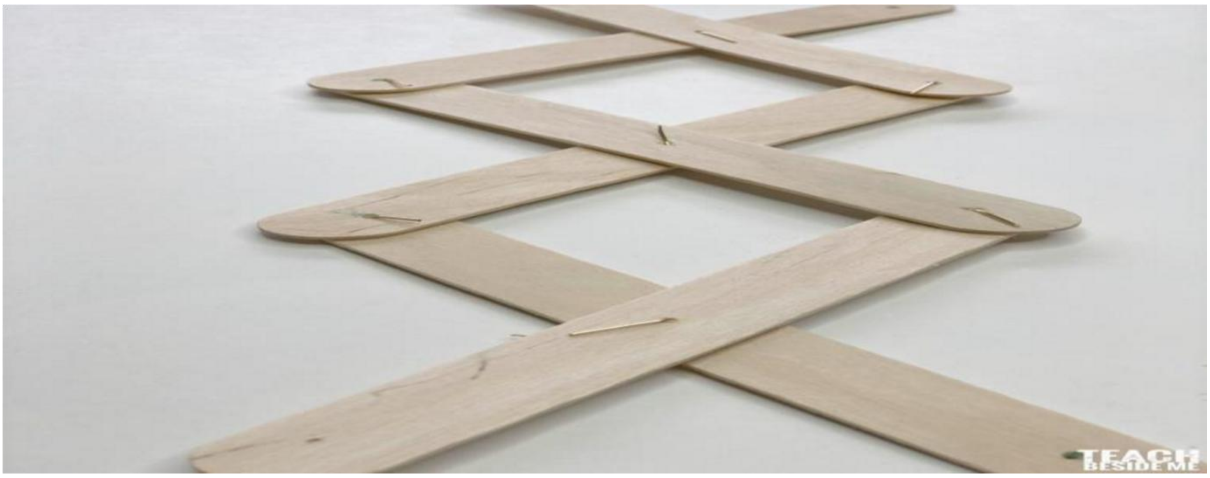
Jumbo Popsicle Sticks (17)
Wire
Wooden Skewers (2)
2 – 10 ml Syringes
Thin plastic tubing

Mark the center and two end points on each popsicle stick (about 1/2 inch in). You will be putting a hole through these points, and you want to them to match up with the other ones. Stack the sticks together and tape them together with some masking tape.

Use a small drill bit and drill through the tree spots. However, it tends to split the stick on the ends a bit, so be careful! Or just use a small screwdriver instead- it takes a little longer, though.



Once the sticks all have holes, connect the center point of two sticks with a small piece of wire. Do this three times. Then connect the two ends of each pair so you have a row of 3 pairs of overlapping sticks.



Repeat with the other half so you have two moving pieces with 6 sticks in each. On the bottom and top holes, combine the two sides together by putting a wooden skewer through the holes. I used 2 skewers cut in half for this step. I also ended up adding a dab of hot glue where each of the sticks were secured to keep them from slipping out.

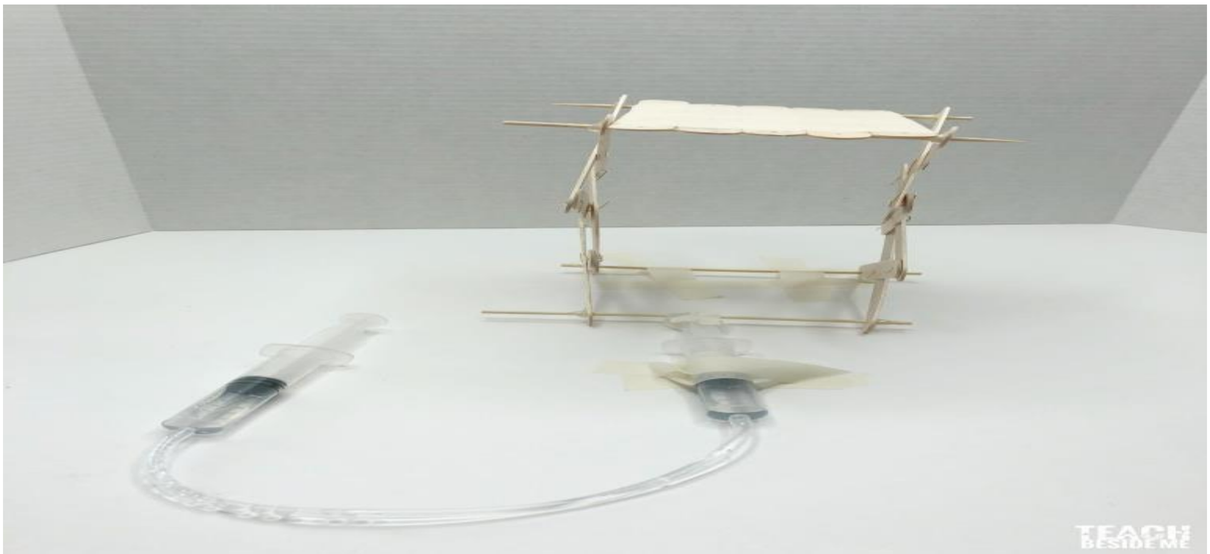
As you can see in the photos, I had a few of my sticks split a bit and I taped them with some masking tape.

If you want to make a platform for the top of the hydraulic elevator, just tape together 5 more jumbo sticks.

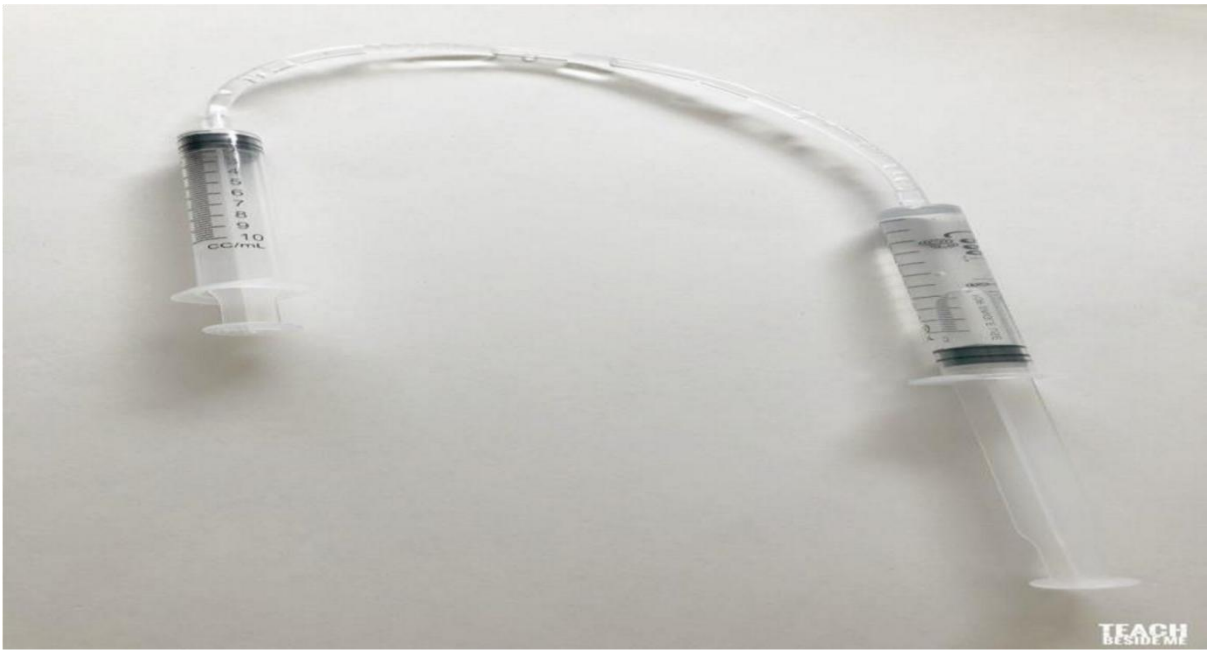
Next you will need to secure the bottom back skewer to the surface you are using. I taped it down to the table to keep it in place.

As you can see in the photos, I had a few of my sticks split a bit and I taped them with some masking tape.
If you want to make a platform for the top of the hydraulic elevator, just tape together 5 more jumbo sticks.

Next you will need to secure the bottom back skewer to the surface you are using. I taped it down to the table to keep it in place.

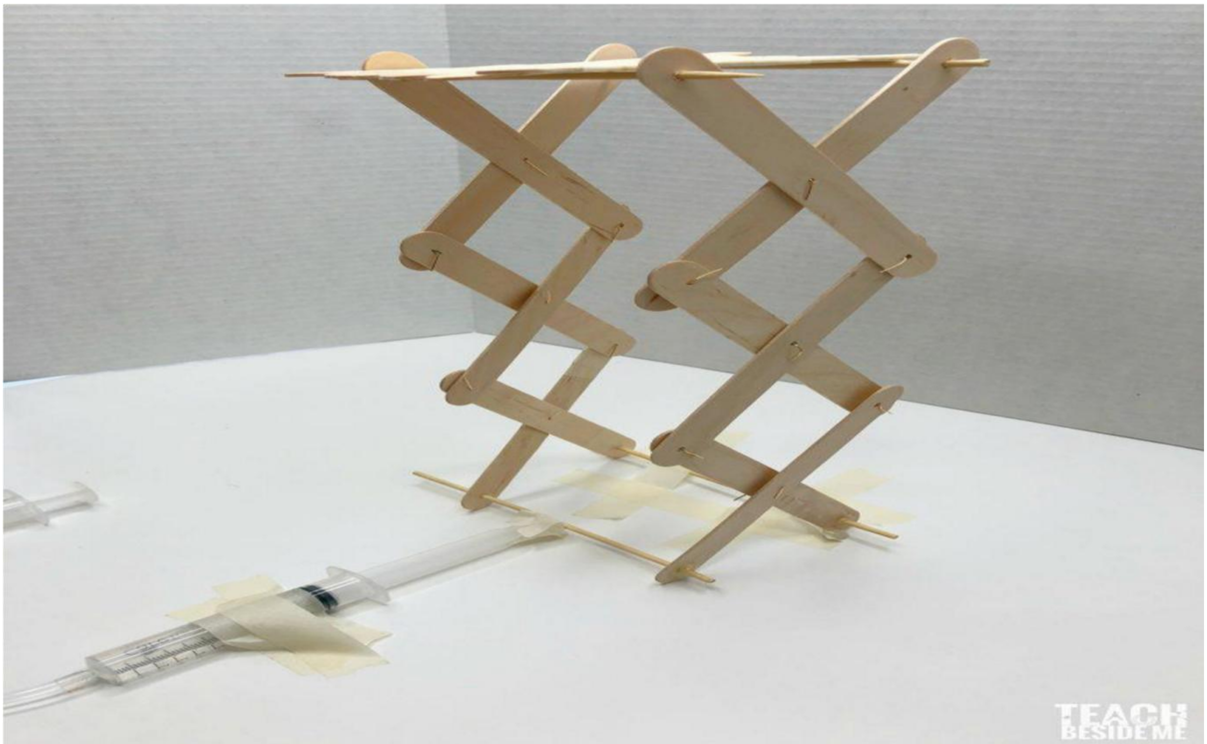


Get your syringes ready now by cutting a small piece of tubing and attaching it to one tip. Fill the other with water and attach it to the other end of the tube.



Once the syringe is prepared, tape one end down to the table as well. The end that slides in and out should be taped to the front skewer.

Now when you push the syringe in and out it will lift and lower the hydraulic elevator.



Experiment-7

AIM

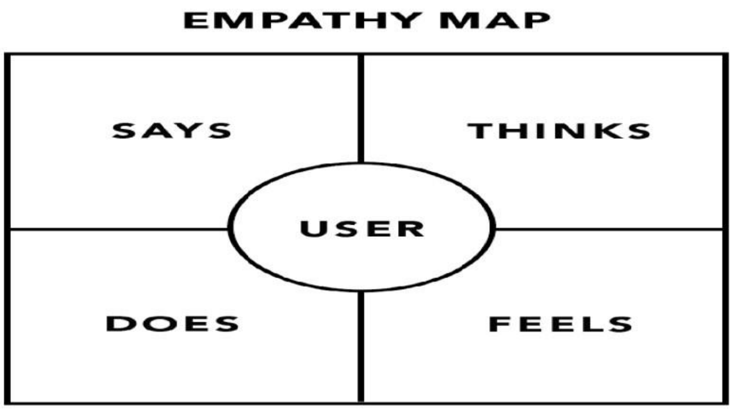
Construct empathy maps for a given case study-1

Description

Definition: An **empathy map** is a collaborative visualization used to articulate what we know about a particular type of user. It externalizes knowledge about users in order to
1) create a shared understanding of user needs, and 2) aid
in decision making.

Format

Traditional empathy maps are split into 4 quadrants (*Says*, *Thinks*, *Does*, and *Feels*), with the user or persona in the middle. Empathy maps provide a glance into who a user is as a whole and are **not** chronological or sequential.



The **Says** quadrant contains what the user says out loud in an interview or some other usability study. Ideally, it contains verbatim and direct quotes from research.

- "I am allegiant to Delta because I never have a bad experience."
- "I want something reliable."
- "I don't understand what to do from here."

The **Thinks** quadrant captures what the user is thinking throughout the experience. Ask yourself (from the qualitative research gathered): what occupies the user's thoughts? What matters to the user? It is possible to have the same content in both Says and Thinks. However, pay special attention to what users think, but may not be willing to vocalize. Try to understand why they are reluctant to share — are they unsure, self-conscious, polite, or afraid to tell others something?

- "This is really annoying."
- "Am I dumb for not understanding this?"

The **Does** quadrant encloses the actions the user takes. From the research, what does the user physically do? How does the user go about doing it?

- Refreshes page several times.
- Shops around to compare prices.

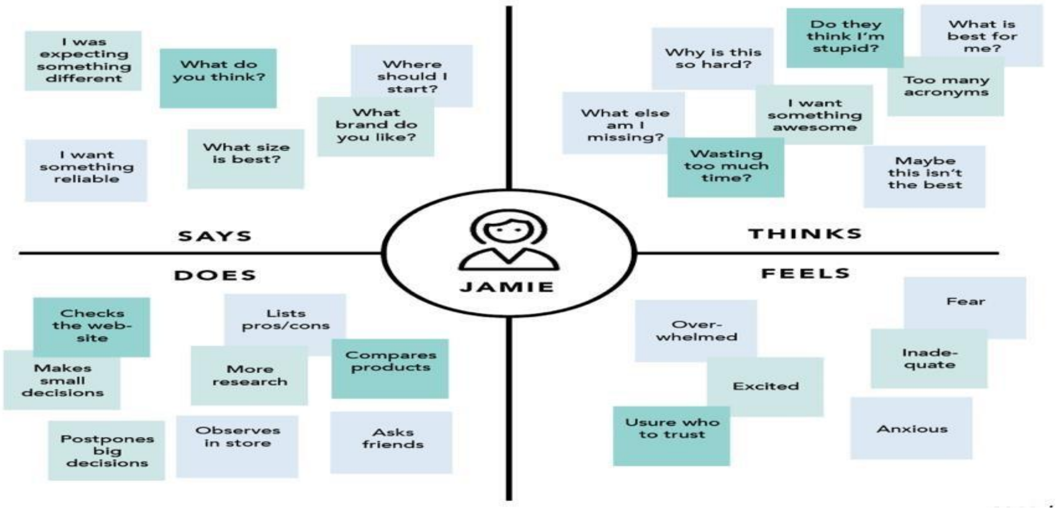
The **Feels** quadrant is the user's emotional state, often represented as an adjective plus a short sentence for context. Ask yourself: what worries the user? What does the user get excited about? How does the user feel about the experience?

- Impatient: pages load too slowly
- Confused: too many contradictory prices
- Worried: they are doing something wrong

Why Use Empathy Maps

Empathy maps should be used throughout any UX process to establish common ground among team members and to understand and prioritize user needs. In usercentered design, empathy maps are best used from the very **beginning of the design process**. Both the process of making an empathy map and the finished artifact have important benefits for the organization:

- **Capture who a user or persona** is. The empathy-mapping process helps distill and categorize your knowledge of the user into one place. It can be used to: ○ Categorize and make sense of qualitative research (research notes, survey answers, user-interview transcripts) ○ Discover gaps in your current knowledge and identify the types of research needed to address it. A sparse empathy map indicates that more research needs to be done. ○ Create personas by aligning and grouping empathy maps covering individual users
- **Communicate a user or persona to others**: An empathy map is a quick, digestible way to illustrate user attitudes and behaviors. Once created, it should act as a source of truth throughout a project and protect it from bias or unfounded assumptions.



- **Collect data directly from the user.** When empathy maps are filled in directly by users, they can act as a secondary data source and represent a starting point for a summary of the user session.

Moreover, the interviewer may glean feelings and thoughts from the interviewee that otherwise would have remained hidden. **Process: How to Build an Empathy Map**

Go through the following steps to create a valid and useful empathy map:

1. Define scope and goals

a. **What user or persona will you map?** Will you map a persona or an individual user? Always start with a 1:1 mapping (1 user/persona per empathy map). This means that, if you have multiple personas, there should be an empathy map for each.

b. **Define your primary purpose for empathy mapping.** Is it to align the team on your user? If so, be sure everyone is present during the empathy-mapping activity. Is it to analyze an interview transcript? If so, set a clear scope and timebox your effort to ensure you have time to map multiple user interviews.

2. Gather materials

Your purpose should dictate the medium you use to create an empathy map. If you will be working with an entire team, have a large whiteboard, sticky notes, and markers readily available. (The outcome will look somewhat like the illustration above.) If empathy mapping alone, create a system that works for you. The easier to share out with the rest of the team, the better.

3. Collect research

Gather the research you will be using to fuel your empathy map. Empathy mapping is a qualitative method, so you will need qualitative inputs: user interviews, field studies, diary studies, listening sessions, or qualitative surveys.

4. Individually generate sticky notes for each quadrant

Once you have research inputs, you can proceed to mapping as a team. In the beginning, everybody should read through the research individually. As each team member digests the data, they can fill out sticky notes that align to the four quadrants. Next, team members can add their notes to the map on the whiteboard.

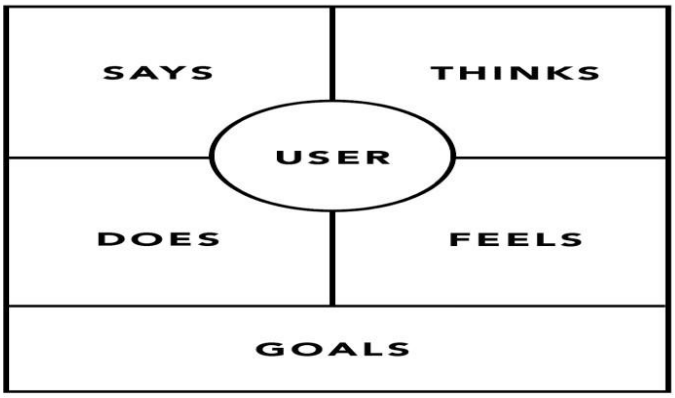
5. Converge to cluster and synthesize

In this step, the team moves through the stickies on the board collaboratively and clusters similar notes that belong to the same quadrant. Name your clusters with themes that represent each group (for example, "validation from others" or "research"). Repeat themes in each quadrant if necessary. The activity of clustering facilitates discussion and alignment — the goal being to arrive at a shared understanding of your user by all team members.

Once your empathy map is clustered, you can begin to vocalize and align as a team on your findings. What outliers (or data points that did not fit in any cluster) are there? What themes were repeated in all the quadrants? What themes only exist in one quadrant? What gaps exist in our understanding?

6. Polish and plan

If you feel that you need more detail or you have unique needs, adapt the map by including additional quadrants (like Goals the example below) or by increasing specificity to existing quadrants. Depending on the purpose of your empathy map, polish and digitize the output accordingly. Be sure to include the user, any outstanding questions, the date and version number. Plan to circle back to the empathy map as more research is gathered or to guide UX decisions.



Conclusion

As their name suggests, empathy maps simply help us build empathy with our end users. When based on real data and when combined with other mapping methods, they can:

- Discover weaknesses in our research
- Uncover user needs that the user themselves may not even be aware of
- Understand what drives users’ behaviors
- Guide us towards meaningful innovation

Experiment-8

Date: ————

AIM

Develop customer journey map for a given case

Description

A *Customer Journey Map* (CJM) is a very helpful tool that represents the whole interaction with a product or service in a transparent manner. It clearly points out the strengths and weaknesses of each stage of the interaction – particularly those that affect the user experience. In addition to this, Customer Journey Maps also show the possibilities for improvement. However, creating a *Customer Journey Map* is a very resource-consuming process. In this article I would like to introduce to you the approach we have taken for one of our clients. The technique that we applied allowed us to quickly create a *Customer Journey Map* in a quick and cost-effective manner.

The Challenge

We were asked to create a *Customer Journey Map* for one of the leading email service providers in Poland. Our client wanted to know why users are choosing their email service, while analyzing the context with which they were using it.

One of the challenges we had to face was to find out what element's users admired the most and the things that should be improved as quickly as possible. Another thing was to identify solutions that would exceed users' expectations and strengthen our client's position against competition. Unfortunately, there were two problems – a tight deadline and budget constraints (I bet this sounds familiar!).

Creating A *Customer Journey Map*

First things first: What input do you need to draw a *Customer Journey Map*? According to Jacek Samsel's article 'Improving UX with customer journey maps', there are 4 main steps in *Customer Journey Map* creation:

1. **Gathering all the information** you already have about your users and the way they use your product or service
2. **Identifying information gaps** and filling them by running additional studies with real users
3. **Creating** *user personas* based on gathered information
4. **Drawing your** *Customer Journey Map* using all this knowledge

However, what should you do in case you do not have time and money for additional *user research* before points 3 and 4?

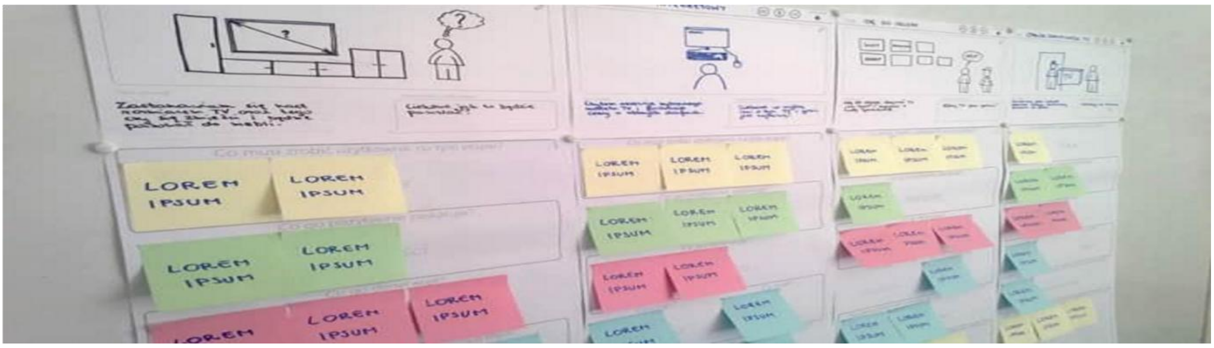
Combining *workshops* & online research

The answer is simple – gather product or service stakeholders, create personas and a **Customer Journey Draft** (CJD). Then verify it via quick online research. We took this approach and it worked out for us:

1. *Workshops*

As mentioned above, the first step consisted of *workshops* carried out with our client. One of the outcomes of these *workshops* was the creation of *User Personas*. Personas are simplified visualization of information about the target group in a form of description of a particular person

Another result of our *workshops* was the Customer Journey Draft – the initial release of the Customer Journey Maps. It was a graphical representation containing a set of hypotheses about users and their interaction with the email service at every stage of the process.



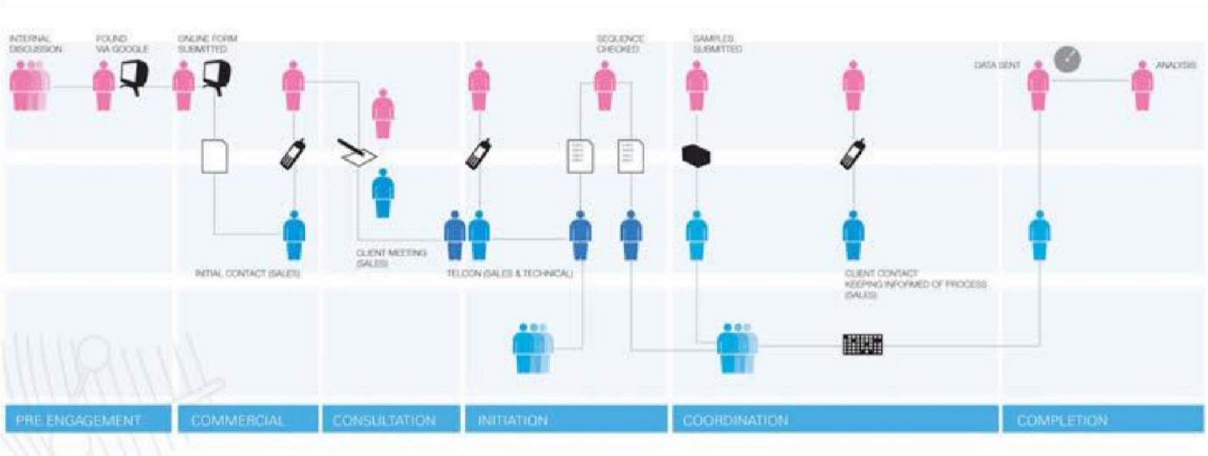
2. Online Research

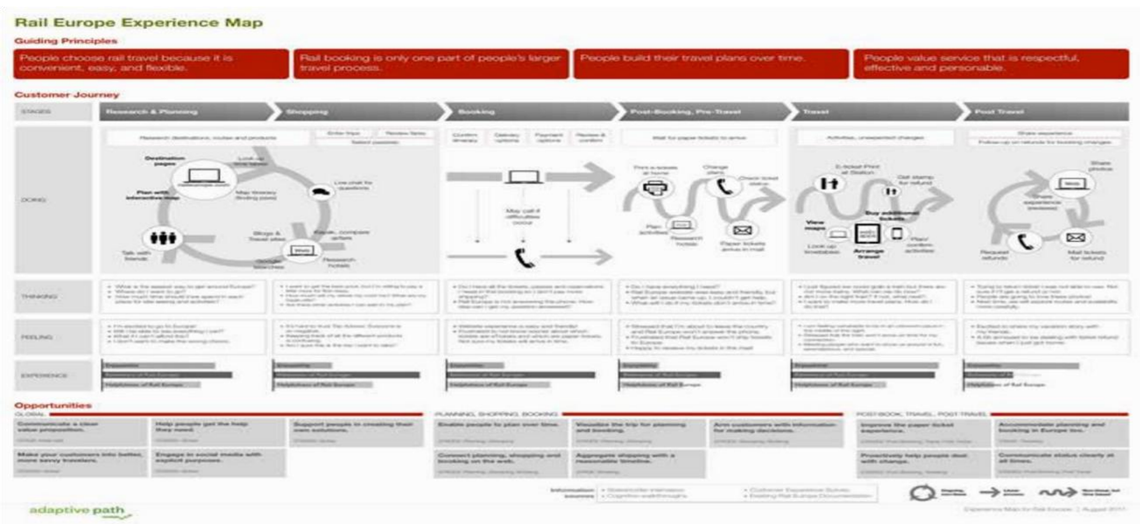
Afterwards, we conducted quick, but detailed research on a about a hundred users using UserZoom. The main goal was to verify quantitatively our hypothesis about Personas and the Customer Journey Draft while extending our knowledge about users. Another purpose of this study was to discover *usability* issues and outline possibilities for improvements. Research also revealed some additional *user needs*, which we addressed so as to exceed their basic expectations.

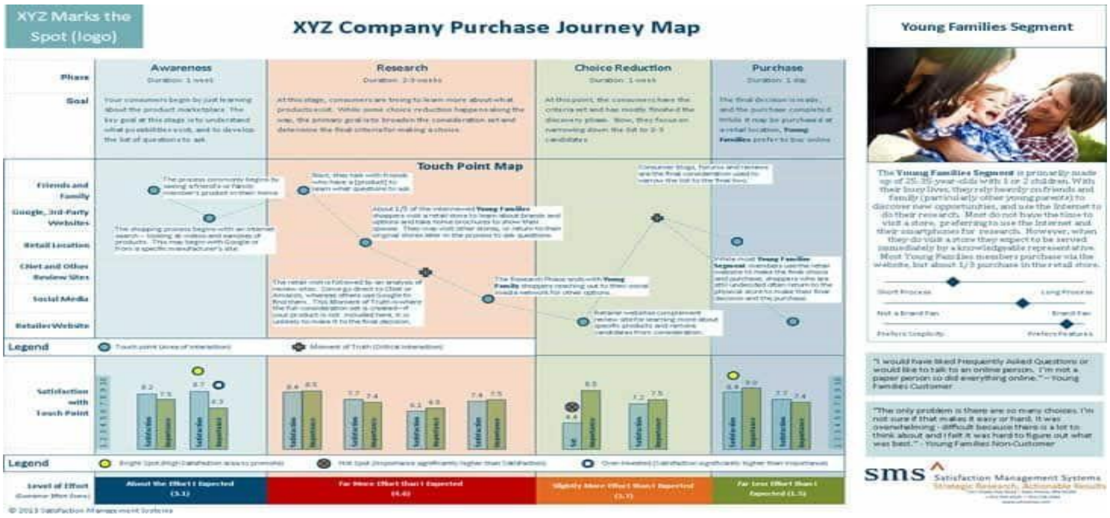
3. Adjusting The Customer Journey Draft

The last step of our approach was adapting the Customer Journey Draft to reflect the research results. We rejected some hypothesis and enriched the Customer

Journey Draft with additional findings. Thanks to that, we achieved the final version of *Customer Journey Map*.







Conclusion

Tight time and cost constraints triggered our innovative approach to modifying the traditional technique of creating a *Customer Journey Map*. Ideally, all the information and insights about users and their interaction with a product or service should be supported by in-depth *user research* during the initial phase.

Experiment-9

AIM

Make a paper prototype for user testing (mock-up model)

Description

Paper prototyping is one of a number of UCD techniques that can be applied early in the development lifecycle to validate or refine a UI design, before time and effort is committed to developing it in software. This article describes our experiences of paper prototyping in the development of Tallis, a clinical knowledge modelling platform developed at the Advanced Computation Lab of Cancer Research UK.

Paper Prototyping

To borrow the words of Carolyn Snyder [2], Paper prototyping is *"a variation of usability testing where representative users perform realistic tasks by interacting with a paper version of the interface that is manipulated by a person 'playing computer,' who doesn't explain how the interface is intended to work".* The advantage of using paper prototyping is not only that it is relatively fast and effective, but also it provides a way to emulate complex logic without having to write any code. Ideal then, for testing an application such as Tallis.

But as always, life is rarely that simple. For Tallis, we had the added problem that in order to convince the relevant stakeholders (and ourselves, for that matter) that this new UI design really was significantly more usable than the old one, and hence worth the investment of 4 months development time, we had to *compare* them side by side. This meant that **both** UIs (old and new) had to be mocked up in paper, even though the first of these already existed in software (well, you *could* compare a design in software with another on paper, but we're not sure that would tell us what we needed to know).

Building the prototype

So we set about the business of converting both UIs into paper. We already had an outline design of our new UI on paper, in the form of PowerPoint, so the initial plan was to re-purpose that so that the various widgets and dialogue boxes could be rendered separately, then printed out on paper and cut out. However, this wasn't as trivial as it sounds - considerable time had to be spent on deciding exactly *what* components would be needed, and in what form (e.g. some dialogue boxes are unchanging, whereas others have content that changes according to context. These need to be "templated" wherever possible, to minimise the number of separate interface components needed for the test itself). Examples of some of the Tallis UI components can be seen in Figure 1.

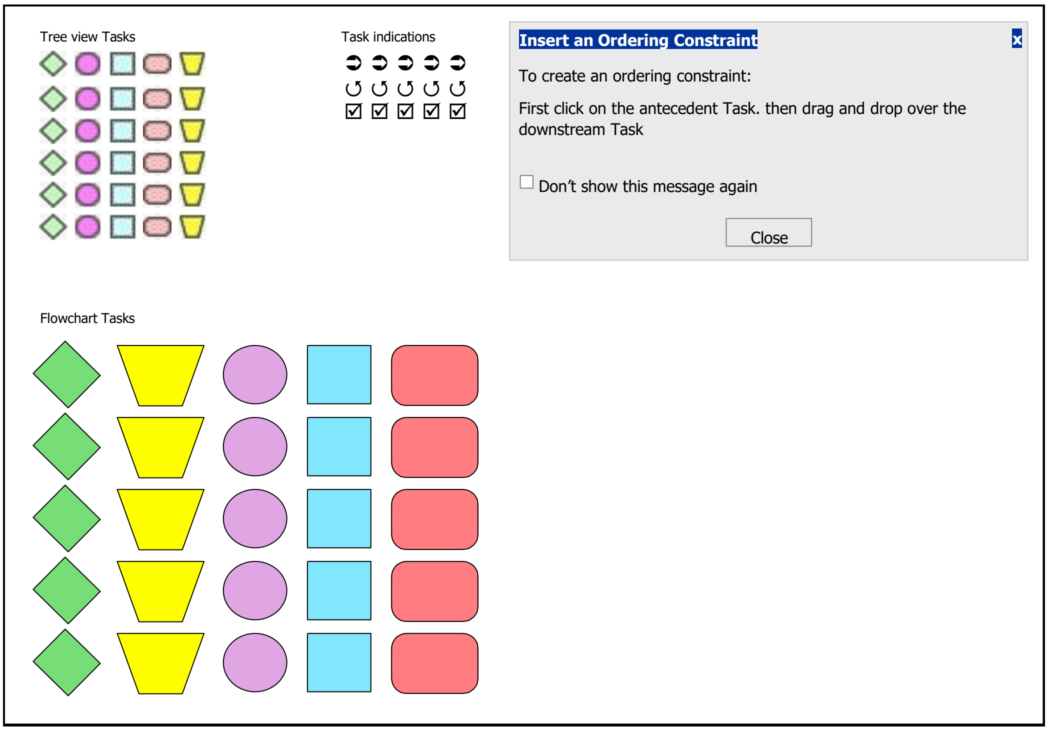


Figure 1: Some of the Tallis UI components

In addition, Tallis was a large and complex application, with many different aspects to its functionality, so it clearly was neither necessary nor desirable to implement the whole of its breadth and depth on paper. Instead, we needed a subset that mapped onto the tasks we intended to give our test participants – hence the need to consider the test script itself.

Participants and test scripts

But before we could think about the test script, we needed some users. We quickly ruled out the possibility of recruiting busy clinicians (our ideal audience), partly because we knew their time was limited, and partly because we needed to prioritise their involvement wherever possible in favour of more the formal scientific work going on in the lab. Instead, we needed an audience who would be interested in what we were doing; yet understanding that this was in some senses a pilot of the methodology itself. We found just such an audience in UCL's

HCI students (to whom we are very grateful for their time and enthusiasm in taking part in this study).

Of course, there's a big difference between HCI students and the clinical professionals for whom this software is primarily intended. But Tallis isn't aimed exclusively at clinicians; (in fact, many users come from a more AI-oriented, "knowledge engineering" background), and from a usability point of view, we felt that many of the key problems with the existing Tallis UI would be experienced by *anyone*, regardless of background. It did, of course, mean that the participants would need to receive a brief lesson in the basics of knowledge modelling, so that when they undertook the test, they would at least understand the high-level principles of what they were had been asked to do, if not the precise UI practicalities.

So, now that we had nailed down our users (metaphorically speaking – although we did at times wonder what to do if one of them chose to leave early) we needed to decide just how much breadth and depth to put in our prototypes. And for that, we needed to return to the issue of tasks.

Choosing which tasks to put in a usability test is always something of a compromise (between time, resources, budget, the issues you wish to explore, and so on), which is well covered in the usability literature (e.g. Rubin [3]), so we won't spend additional time on it here. Suffice it to say that we identified half a dozen tasks, ranging in complexity, which we thought would exercise the majority of the key problems with the existing UI in a fixed 90-minute period. We also chose those tasks on the basis of their suitability for some sort of quantitative analysis (e.g. error rates, completion times, etc.), so that we could eventually derive a set of numerical metrics with which to compare the two UIs. We then documented these as a task script, using suitably neutral terminology (so as not to present a bias in favour of any one interface). We also set a maximum time in which to complete each task, after which we would simply move on to the next one.

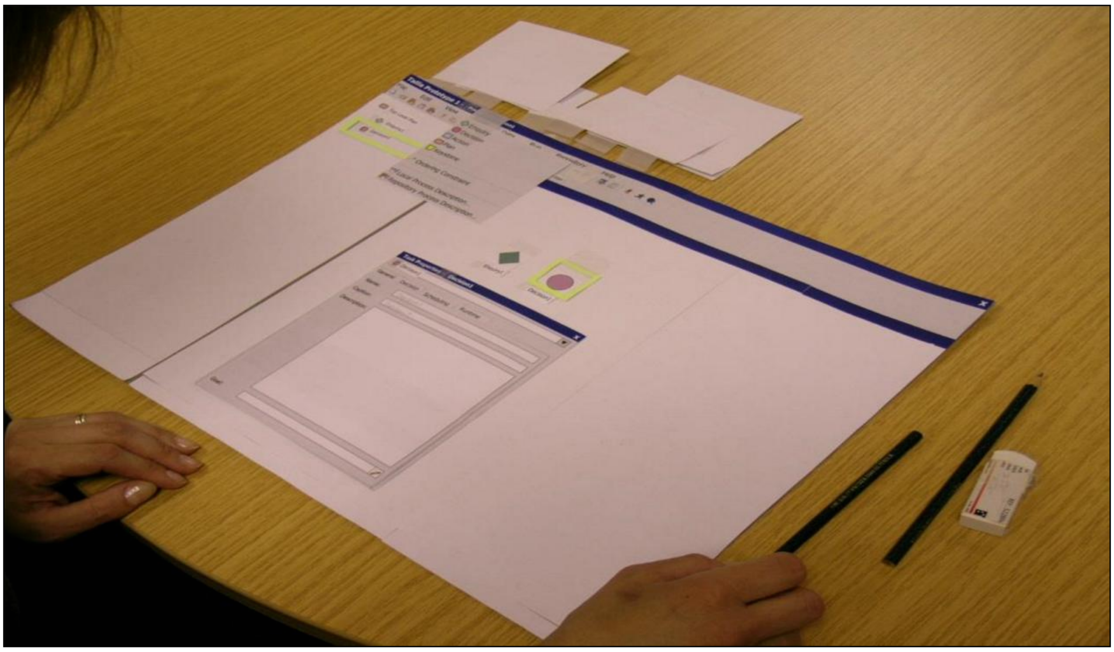
Finally, we had to prepare also a paper prototype of the existing UI, by creating a further PowerPoint mockup (so that the look and feel would be consistent with the first), then repeating our game of print – paper – scissors. In all, it took us probably 3 weeks to prepare the paper prototypes, which in hindsight was possibly a little indulgent. However, in this period we did mock-up **two** separate

UIs, and we did experiment with a lot of variations on the prototyping process. And it was a highly complex UI (as visual authoring UIs tend to be, with many components that can be arranged and manipulated on a 2D canvas).

The Test Itself

Our test involved eight participants (we did originally hope to recruit more), which we stratified into two groups (one for each UI) and also by gender/background. The test itself was heavily scripted, so as to minimise any variation in our behaviour that might otherwise bias the comparison. The experimental setup is shown in Figure 2, in which a participant can be seen interacting with the paper prototype. Just out of shot are the moderator (the individual running the session) and the Computer (who operated the prototype).

As is typical with paper prototyping, there is no keyboard or mouse – to enter text you just use the pencil, and for mouse movements you just use your finger (and say whether you are right clicking, hovering, etc.). The drop-down menus can be seen at the top of the picture, attached to the main application window with sticky tape and are revealed or concealed as & when appropriate.



The Results

Our test was primarily designed to elicit quantitative measures with which we could compare the usability of the two UIs. However, we won't be going into any detail with the results here, as (with only eight participants) they're not significant, and besides, they are not really the focus of this article anyway. Moreover, as is often the case with usability testing, the most valuable results can be the qualitative (verbal) feedback received during the test itself. Nonetheless, Figure 3 shows an example of the quantitative results: the number of participants to fully complete each task for each UI. As can be seen, the redesigned UI (UI2) performs better in almost every case.

Similarly, the redesigned UI fared better on just about every other metric: levels of completion (of each task), average task score, time to complete tasks, etc. But in a way, the detailed results aren't really that interesting for an article on paper prototyping – what matters is the methodology we used: to what extent that was a success, what we learnt from it, and whether we'd use it again.

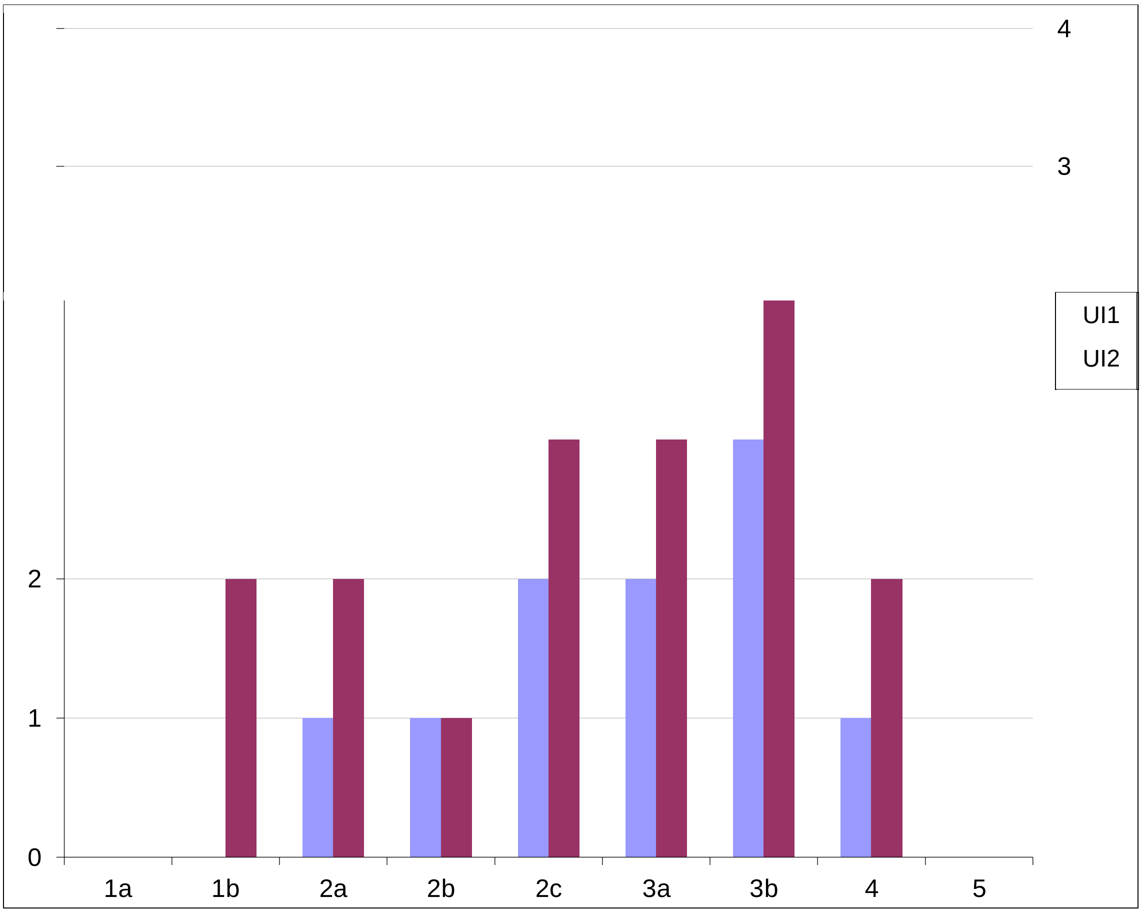


Figure 1: Number of Participants to Fully Complete Each Task

Conclusions

So, was paper prototyping a success? Almost certainly. It allowed us to explore the differences between two complex UIs and find answers to some difficult questions within a relatively short space of time. In fact, one of the good things about paper prototyping is that it allows you a certain amount of opportunism – for example, we could see by half way thru the test that participants were struggling to find a certain menu item – so we modified the menu structure, and thus solved the problem for later participants. Obviously, there is a judgement to be made as to precisely when to make such changes (they can inevitably affect the validity of the experiment in a scientific sense), but on this occasion it seemed more important to take the opportunity to test a potential solution to a known problem.

Experiment-10

AIM

Design and development of cell phone wallet (mock-up model)

Description

The Mobile Wallet can be understood as the mobile version of the ewallet. The electronic wallet on smartphones is reshaping the process of purchasing goods and services, either remotely or in close proximity, and they are protagonists in the new generation of digital payments.

There are several major international players who have focused strongly on the payment technologies of Mobile Wallets, and there are even more users in the world who use this innovative tool daily to finalize their payments. And it is from the users themselves that one can understand how to improve this technology.

What is meant by Mobile Wallet - functions and types:

When we talk about Mobile Wallet we mean an app for smartphones, which aims to replace the physical wallet with a digital one. Various payment instruments may be contained within it, but also numerous services that the user can access close to or remotely. A Mobile Wallet can, therefore, allow you to:

- pay, for example by bringing the phone closer to a contactless POS;
- contain and enjoy discount coupons;
- virtualize loyalty cards or travel tickets;
- send money to a friend;
- contain identity documents (at least potentially).

There are several approaches to developing Mobile Wallets: there are feature-rich wallets (such as AliPay or PayTM); those mainly specialized in payment (such as Samsung Pay or Apple Pay); vertical ones, such as loyalty card aggregators (e.g. Stocard) or that leverage cashback and loyalty programs (e.g. Yoyo Wallet), or that focus on p2p transfers (e.g. Circle).

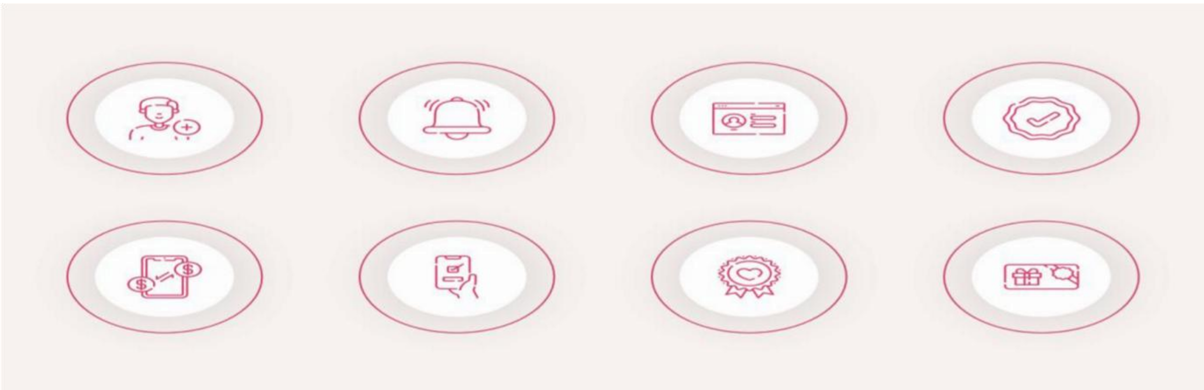
Key Benefits that Mobile Wallet Applications Can Bring

Why should you develop a mobile wallet app?

Such applications can deliver many benefits to businesses in every industry.

- **Accessibility:** Mobile wallets allow making day-to-day transactions without carrying around physical cards. All they need is an app on their smartphone and a user ID.
- **Access to money from mobile:** Users can start making mobile payments in no time. They need to connect their cards to the digital wallet and make payments with smartphones.
- **Variety of use cases:** Mobile wallets can be used for a range of uses. Users can pay for purchases, buy tickets, pay for online services, etc.
- **Bills splitting:** Depending on where you are using the wallet app, you can add the functionality allowing users to split the bill and send a link to users who needs to pay them money.
- **Automatic payments:** You can add the ability to set up automatic payments within your app so users won't forget about them because of their hectic everyday life.
- **Promotions & discounts:** Digital wallets usually offer different promotions and discounts. You can reward users for buying at partners' locations, making a certain number of monthly payments, etc.

Basic & Advanced Features of Mobile Wallet Apps



Don't skip this section if you're interested in creating a digital wallet. It provides examples of basic and advanced functionality that you might need to include in your application.

- **User registration** – registration of a user within your application with different authentication methods, such as mobile phone, email address, social media, etc.

- **Push notifications** – a valuable feature to notify users about successful and declined payments.
- **Banking account authorization** – the ability for users to connect their payment cards to your wallet application.

- **Balance checking** – a feature that allows users to check their account balance right from the mobile wallet.

- **Transactions** – the ability to work with transaction operations (sending and receiving money).

- **Bills payment** – a feature that allows paying different types of bills.

Aside from basic features, you might also want to consider adding some advanced functionality:

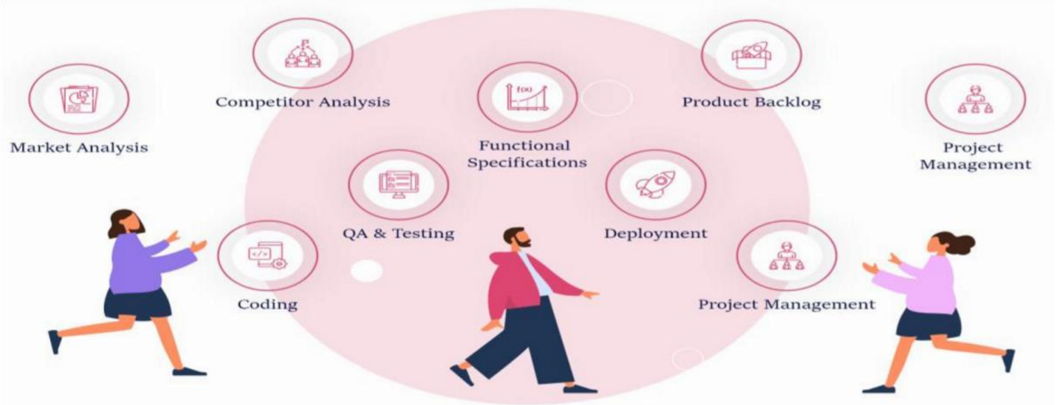
- **Loyalty cards** – the ability to use loyalty cards during shopping or transferring money.

- **Gift cards** – a feature that allows adding gift cards and using them to pay for purchases.

- **Exclusive offers –** notifications about special offers provided by you or your partners.

Mobile Wallet App Development: Services & Processes

The development of wallets for mobile devices is a process comprising different steps. Usually, development is divided into discovery and development stages.



1. **Discovery Stage**

The **discovery stage** is an essential part of app development. During this process, companies identify their business needs, define technical requirements and match them with technologies available on the market.

You cannot skip the discovery stage if you want to create an app with a chance for success.

At the end of this stage, you will have the following documents on your hands:

- Market analysis
- Competitor analysis
- Functional specifications
- Product backlog

The lion's share of the discovery stage is dedicated to UX and UI design. The deliverables of this step include UX wireframes, UI mockups, branding elements, illustrations, animations, etc. During the wallet app development, you will need to identify how users will interact with your solutions and what it will look like for the end-users.

2. **Development Stage**

During the **development stage**, the coding itself happens. Developers create the code base of your application, integrate it with third-party solutions, and implement custom features.

The development stage is traditionally divided into several phrases:

- **Coding** – code base creation, integration with SDKs, APIs, etc.
- **QA & Testing** – thorough application testing to identify bugs that might interfere with the app's use.
- **Deployment** – release of the developed app to the market.
- **Ongoing development & Support** – working on further app iterations, updating the app with new features and platform support.

How to Make a Mobile Wallet App: Step-by-Step Guide

The development process of a mobile wallet starts long before coding. The project implementation is a complicated process comprising multiple stages, each having its crucial role in the whole process.

1. Market Research

App development usually starts with market research and competitor analysis. During this stage, you need to decide for what industry you will develop an app, who your target audience is, and who your competitors are.

Market research will help you conceptualize your application and find your value proposition.

2. App Concept Finalization

This stage is crucial for the future of your application. You need to finalize the concept of your future apps. In simple words, you need to decide on the USP (Unique Selling Proposition) and how to set your app apart from the existing competitors.

3. Choosing App Development Company

If you don't have the necessary technical knowledge, you will need to find a development company that will create a wallet for you. Here are a few tips on how to choose a company that will become your reliable technical partner:

- Start the search online on Clutch, ITFirms, Manifest, etc.
- Check portfolio and customer reviews
- Conduct interviews with vendors
- Ask about relevant experience and similar projects

During the screening process, some aspects can influence future cooperation, such as the level of English knowledge, differences in time zones, cultures, etc.

4. Custom UX/UI Design

The interface creates the first impression of a mobile page. What users see first when they open a new app decides whether they will use it or abandon it right away. Thus, you need to address this part of all development with an eye for detail. The user interface should be feature-rich, user-friendly, and attractive.

5. **Mobile Wallet App Development**

Developers then turn your UX wireframes and UI mockups into a working application and implement all the necessary features.

Remember that mobile wallet applications should be created with scalability and security in mind.

6. **App Launch & Promotion**

After you launch the app to the market, it’s time to think about its promotion. In most cases, marketing strategy creation starts long before the app is released to the end-users.

Today, a comprehensive marketing strategy includes several parts, such as SMM, content marketing, paid ads, etc.

7. **Ongoing Development & Support**

The development process doesn’t stop at the previous stage. If you create an app with a long-term outlook, you will need to continue working on it.

After getting the first user feedback, you might want to add some functionality, improve existing features, and plan for further updates. The process of app development is ongoing.

Conclusion

Mobile wallets are no longer just a trend they have become a need of the time.

However, to relish the benefits of this fast-evolving market, you need to ensure flawless implementation of facts and additional features.

Do the necessary market research to build and app for mobile payment solution.